



Full length article

The resonant actuation and optimization design of piezoelectrically actuated bistable composite laminates

Yuting Liu¹, Ke Huang¹, Jiaying Zhang^{*}

School of Aeronautic Science and Engineering, Beihang University, Beijing 100190, China

ARTICLE INFO

Keywords:

Bistable
Macro-fiber composite
Resonant actuation
Response surface methodology (RSM)
Pareto front

ABSTRACT

Asymmetric cross-ply laminates exhibit two stable configurations due to thermal residual stresses developed during or after curing and cooling process, which may cause them to bend in both directions. Bistable laminates, such as asymmetric cross-ply laminates, exhibit excellent response speed and load-bearing capacity, and strong nonlinear and negative stiffness characteristics during deformation. These characteristics make bistable laminates highly promising for wide applications in morphing wings, soft actuators, energy harvesting and electronic devices. Especially in the aerospace field, morphing wings composed of bistable laminates can improve flight performance, enhance control, and reduce drag and flutter. To overcome the shortcomings of traditional discrete rigid deformation structures and promote the development of adaptive structures to lightweight, self-driven and high-performance systems, this paper integrates Macro Fiber Composite (MFC) into bistable laminates to replace the traditional exciter for actuation. An analytical model is derived using Hamilton's principle and the Rayleigh-Ritz method, by comparing the calculation results of commonly used fourth-order and sixth-order dynamic analytical models with ABAQUS finite element shell models to determine systematic parameter and stable configurations. Summarizing the advantages of the existing dynamic actuation, this paper proposes resonant actuation which can change the structure to ensure that the inertia forces and stiffness forces cancel each other over the desired actuation period. Based on the strong nonlinear characteristics of bistable laminates, the method for determining the critical actuation load is systematically summarized for the first time. In order to find bistable laminates with low energy actuation and large deformation, taking the size of the composite laminates as a variable to establish a global mathematical model of critical actuation load by Response Surface Methodology (RSM). Taking the actuation voltage required for one unit deflection as the objective function, the non-dominated sorting genetic algorithm-II (NSGA-II) is used to find the Pareto front, determining the best operating specimens for experimental verification. This approach lays a theoretical foundation for exploring low energy consumption paths and global optimization of MFC actuation, paving the way for the large-scale application of morphing wings.

1. Introduction

Traditional deformable structures require continuous external energy input or locking mechanisms to maintain their configuration after deformation [1]. Thus, it often demands for additional control devices and load maintenance devices in the system, which will inevitably increase structural complexity and weight. In contrast, bistable laminates embedded with MFC can achieve large deformation through aperiodic oscillation or snap-through with less energy input [2]. This approach ensures the integrity and reliability of the laminates before and after configurations switching. The bistable laminate can function as both

load-bearing structure and highly deformable component. Therefore, bistable composite laminates with morphing behaviour have attracted extensive attentions for various engineering applications [3].

Previous research on asymmetric composite laminates, particularly before the 21st century, primarily focused on configuration prediction and tension-bending coupling effects. In theoretical calculations, the bistable phenomenon of asymmetric laminates was mainly attributed to the von Kármán nonlinearity and geometric parameters of the laminates, such as the aspect ratio and length-to-thickness ratio [4]. Hyer, the discoverer and proposer of bistable laminates, has made significant contributions to subsequent theoretical research and practical applica-

^{*} Corresponding author.

E-mail address: jiaying.zhang@buaa.edu.cn (J. Zhang).

¹ Contribute equally to the manuscript.

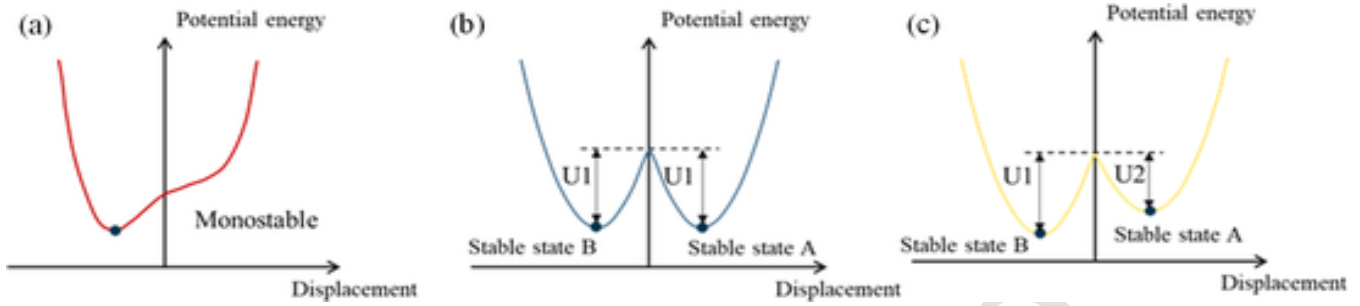


Fig. 1. Potential energy diagram of asymmetric laminate, (a) monostable, (b) symmetric bistable, (c) asymmetric bistable.

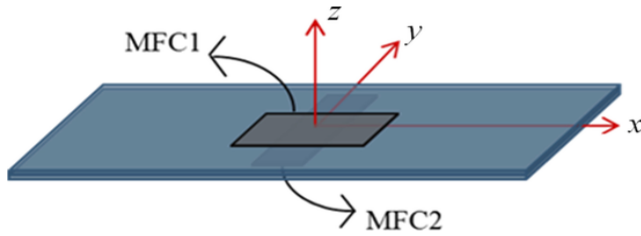


Fig. 2. Bistable laminates with orthogonal lay-up MFCs.

tions [5,6]. Hyer first demonstrated that Classical Lamination Theory is not applicable to asymmetric composite laminates, since such laminates undergo significant deformations and pronounced bending after curing. This behaviour contradicts the fundamental assumptions of linear elasticity and small-deformation kinematics inherent in the Kirchhoff hypothesis. Hyer derived a polynomial displacement field by assuming that the laminate is positively cylindrical after curing and cooling, so that the principal curvature is constant. Using the Rayleigh-Ritz method based on the principle of minimum potential energy, he found the local extrema of the energy equation. The results in [7] indicated that only terms whose powers of x and y add up to even numbers exist in the displacement field and strain fields of laminates with arbitrary ply angles. Therefore, by neglecting higher-order terms, a simplified 11-parameter polynomial analytical predictive model can be obtained.

In 1987, Akira and Hyer et al. [8] used a theoretical model with a positive cylinder shape to study the relationship between the curvature and curing temperature of $[0^4/90^4]$ laminates. Their results showed that the bending curvature of the laminate would exhibit a bifurcation phenomenon as the temperature changes. The theoretical model that did not consider the change in curvature led to a large error compared to the experimental results in determining the structural bistability. Subsequently, Jun and Hong et al. [9] conducted research on bistable laminates based on Hyer's work. They proposed that the laminate at room temperature was not a perfect cylinder due to the phenomenon of anti-elastic curvature near the edge, and the in-plane shear strain could not be ignored. Later, they reduced the 11 independent parameters to 7 [10]. Schlecht and Hyer et al. [11] compared the predicted results of the four-parameter analytical model with finite element results. The comparison showed that, although the analytical model greatly improved computational efficiency, it was unable to predict the free edge effect. Peeters et al. [12] conducted experimental measurements on orthogonally and arbitrarily oriented asymmetric composite laminates and compared the results with the analytical model. They analyzed the influence of manufacturing parameters and other factors on the geometric errors. In 2007, Mattioni et al. [13] proposed the concept of segmented bistable laminates and established a 33-parameter configuration prediction model for segmented laminates, assuming different displacement function forms for different regions within the laminate. In

2018, Wu et al. [14] proposed a 17-parameter configuration prediction model for bistable laminates, in which the coupling between the bistable laminate and the shaker and the effect of stretching-bending coupling on the mid-plane strains are fully considered. The nonlinear characteristics of single-well oscillations and cross-well oscillations were predicted. Pandey et al. [15] introduced Kirchhoff thin plate theory into a two-dimensional finite element model, compared it with a three-dimensional finite element model based on third-order shear deformation theory in ABAQUS, and thoroughly discussed the convergence and stress analysis of smart piezoelectric composite plate and shell structures.

Subsequent in-depth research revealed that the predictive accuracy of configurations cannot be improved by changing the deformation assumptions or increasing the degree of freedom of the displacement and strain field functions. The research focus of scholars began to turn to configuration switching of bistable laminates—snap-through, as well as the corresponding actuation methods, actuation paths, actuation materials, and optimization of variables according to different actuation objectives. Dano and Hyer [16,17] simulated the snap-through behavior of bistable laminates using two reverse torques to mimic the form of shape-memory alloys (SMAs), and their results closely matched the critical actuation force predicted by the experiments. They ultimately demonstrated that using multi-parallel SMAs is an effective way to achieve snap-through. Pirondi et al. [18] embedded SMA wires into laminates to achieve configuration switching. However, despite the large deformation capabilities of SMAs, their response is slow, making them unsuitable for rapidly changing dynamic environments in aerospace applications. As a result, Schulte and Hyer [19,20] began to use another type of smart materials—MFC, to achieve snap-through. MFC actuators are easy to integrate, lightweight, responsive, and highly flexible, but possess low strain with a maximum achievable strain of 0.2 %. Therefore, it is necessary to firmly adhere the MFC to the surface of the bistable laminate to ensure the transfer of strain. They obtained the critical snap-through voltage through theoretical calculations, finite element simulations, and experiments. However, the theoretical predictions deviated significantly from the experimental measurements. The average snap-through voltage in the experiments exceeded 1500 V, which was beyond the MFC's working voltage limit.

Schultz and Wilkie [21] pasted MFC actuators on both sides of $[0/90]$ bistable laminates to achieve two-way actuation. The MFC actuators altered the characteristics of the bistable laminates, resulting in a multi-stable structure with transitional states during actuation, making it unable to achieve synchronous snap-through. Bowen et al. [22] investigated the experimental actuation of bistable beams in a $[MFC/0^2/90^2]$ layup. The experimental results showed that the cantilever beams could only achieve one-way snap-through under piezoelectric actuation when a certain concentrated force was applied at the end of the cantilever beams. Portela et al. [23] modified the thermal expansion coefficient by considering moisture absorption of composite materials. They suggested that for a given bistable laminate, a smaller MFC could not gen-

Table 1
Material parameters of T700 carbon fiber reinforced prepreg and MFC.

	Symbol	Quantity	Value
T700	E_1	Axial tensile modulus	124.9 GPa
	E_2	Transversal tensile modulus	7.9 GPa
	G_{12}	Shear modulus	5.6 GPa
	G_{23}	Shear modulus	5.6 GPa
	ν_{12}	Poisson's ratio	0.3
	α_{11}	Longitudinal thermal expansion coefficient	$-0.1 \times 10^{-6} \text{ } ^\circ\text{C}^{-1}$
MFC	α_{22}	Transverse thermal expansion coefficient	$2.7 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$
	E_1	Axial tensile modulus	28.6 GPa
	E_2	Transversal tensile modulus	13.1 GPa
	G_{12}	Shear modulus	4.03 GPa
	G_{23}	Shear modulus	4.03 GPa
	ν_{12}	Poisson's ratio	0.29
	d_{33}	Piezoelectric coefficients	$374 \times 10^{-12} \text{ m/V}$
	d_{31}	Piezoelectric coefficients	$-171 \times 10^{-12} \text{ m/V}$

erate sufficient actuation force, while a larger MFC might cause the structure to lose its bistability. Kim and Betts [24] combined the advantages of SMA and MFC to achieve two-way snap-through of cantilever bistable laminates using a shape memory-piezoelectric active structures in experiments. Lee et al. [25] fabricated a piezoelectric bistable laminate consisting only of MFC, with a layout of $[0^{\text{MFC}}/90^{\text{MFC}}]$. This structure showed greater deformation, lower snap-through requirements and reduced design constraints. Zhang et al. [26] used a simplified spring-mass model to study the inherent nonlinear dynamic characteristics of the structure and utilized SMA to achieve active switching of bistable beams. Anilkumar et al. [27] used distributed MFC patches to overcome the local large curvature and the sharp snap-through of large-sized MFC, which would reduce the snap-through voltage. In previous studies, most loading conditions were symmetric. Deshpande et al. [28] investigated asymmetric loading of bistable laminates, highlighting the importance of curvature changes during the snap-through process, and the best actuation position can be selected according to the curvature change characteristics.

Due to the high external driving force required for quasi-static actuation, which is greater than the critical force of the bistable configuration switching, a high-power actuation device is required. In order to realize low-energy consumption actuation, research has gradually turned to the dynamic actuation of bistable laminates. Casals-Terre et al. [29] first proposed the dynamic actuation of bistable structures to achieve snap-through, by applying a harmonic loading of a certain frequency to rapidly switch. Diaconu et al. [30] studied the effects of inertia and damping terms on the snap-through of bistable structures. The analytical model can be calibrated with the Rayleigh damping param-

eters α and β (α is Mass-proportional, β is Stiffness-proportional damping) to match finite element and experimental results. Senba et al. [31] proposed the concept of the MFC vibration actuation. In the resonance region of the bistable laminates, the maximum displacement of the structure is increased by applying a harmonic voltage to the MFC. When the excitation amplitude and frequency of the laminates reach a certain value, the snap-through can be realized. Wang et al. [32,33] focused on the evolution laws of the dynamic characteristics of bistable laminated plates under the changes of excitation amplitude and frequency. Arrieta et al. [34] proposed a dynamic actuation based on modal frequency, achieving configuration control of a cantilever bistable laminate under harmonic loading through MFC. Bilgen and Arrieta [35–37] designed a bistable piezoelectric composite wing and conducted aerodynamic performance experiments. A pneumatic evaluation method for piezoelectric bistable laminate dynamic control technology is proposed. The dynamic response of bistable airfoils under aerodynamic loading is used to make the structure snap-through under lower motivation voltage. Rouleau et al. [38] used a shaker to apply vibration shock loads to a single-degree-of-freedom bistable system and derived analytical equations to estimate and predict the excitation amplitudes corresponding to single-well motion and cross-well motion. Wu et al. [39] obtained the stable configuration of the four-corner simply supported bistable composite laminates by analytical model and finite element model and studied its snap-through and chaotic oscillations laws using the maximum Lyapunov index, bifurcation diagram, three-dimensional phase diagram and Poincaré section.

Owing to their bistability, strong nonlinearity, dynamic stability, and low-energy actuation, bistable laminated plates can be widely applied in various fields, such as morphing aircraft mechanisms, deployable space structures, soft robots, energy harvesting, vibration reduction and buffering, intelligent structures, and so on. Mattioni et al. [13] installed a bistable plate at the tip of a traditional wing to act as a winglet. In its flat configuration, the plate extended the wingspan to augment lift. In its bent state, it reduced the local angle of attack, promoting airflow separation at the wingtip to optimize cruise performance. Rader et al. [40] used piezoelectric actuators for vibration control of flexible aircraft wings, using Genetic Algorithms to find the optimal position for a single actuator. Diaconu et al. [41] utilized bistable laminates to change the shape of the airfoil, proposing three deformation mechanisms to change the wings' trailing edge, camber, and cross-sectional chord length, respectively. Panesar et al. [42] manufactured a bistable laminate with variable angle tows for a morphing flap application and used the Ant Colony optimization with out-of-plane displacement as the fitness function for optimization. Betts et al. [43] aimed to design a structure with both large deformation and high load capacity,

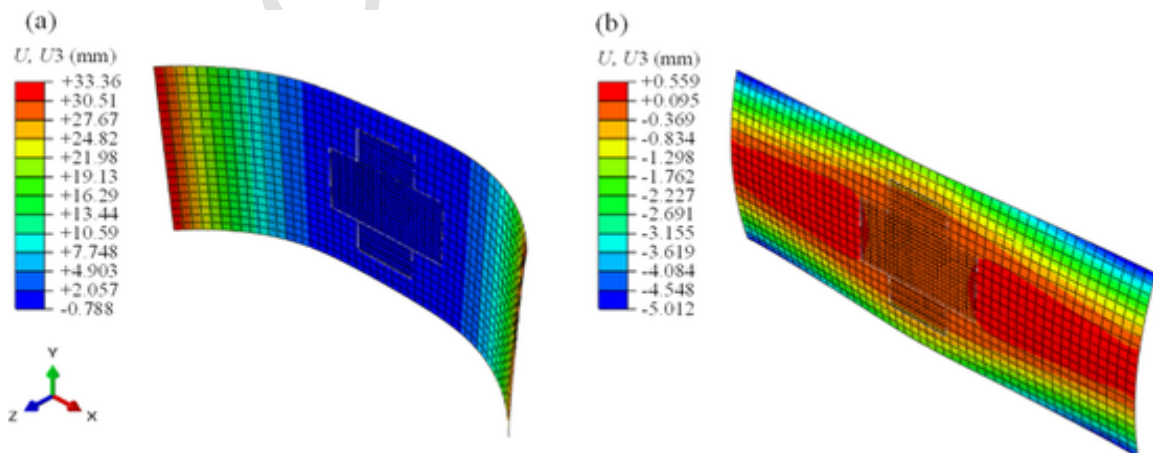


Fig. 3. The out-of-plane displacement landscapes of stable configurations, (a) stable state A, (b) stable state B.

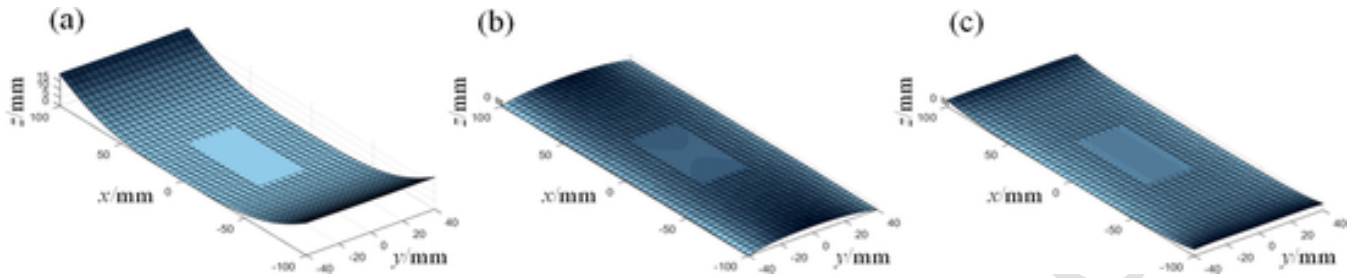


Fig. 4. Stable configuration predicted by analytical model, (a) stable state A, (b) stable state B, (c) unstable state.

so making the bending stiffness maximum in the load direction and minimum in the snap-through direction by designing the thickness and size of the layers. On this basis, two MFCs were arranged as the actuators in an orthogonal way to find the lowest actuation total voltage. Nicassio et al. [44] proposed a bistable composite material morphing surface based on the “lever effect”, which could be integrated into the wing flap of an aircraft to achieve lightweight, passive deformation, and high efficiency. Under certain boundary conditions and external environments, this structure can convert aerodynamic loads in the environment into actuating forces. Inspired by the Venus flytrap, Zhang et al. [45] developed a novel robotic gripper using bistable cylindrical shells as flexible fingers and controlled the configuration transformation using non-contact magnetic actuation. Wang et al. [46] produced a similar biomimetic bistable steel shell device, with two 3D-printed air chambers as the actuators. By inflating the air chambers, non-contact, autonomous, and high output grabbing was achieved. Lu et al. [47] designed a bistable piezoelectric composite laminate with both vibration isolation and energy harvesting functions. Vasios et al. [48] arranged bistable circular shells successively to form a multi-stable structure and controlled the mechanical signal propagation by controlling the snap-through of each bistable shell. This structure could be used for shape reconstruction, rapid deployment, and mechanical logic gates. Dong et al. [49] proposed a novel morphing airfoil structure based on a four-corner, simply-supported bistable composite laminated shell to support aircraft morphing assemblies. In a further study [50], they investigated the intra-well and cross-well dynamics of bistable laminates, revealing that increasing the number of layers enhances the actuation amplitude and broadens the frequency range for dynamic snap-through motions. Bashir et al. [51] identify a suitable size, location, and distribution pattern of the MFC patch to induce minimal asymmetry in the properties of the laminate-MFC assembly. These results can be utilized in the design of a BPEH with minimal asymmetry in its potential energy landscape and static and dynamic properties for efficient piezoelectric energy harvesting.

The bistable structure refers to a structure that naturally has two stable configurations. The asymmetric orthotropic composite laminate easily forms a bistable structure due to the existence of residual thermal stresses after curing and cooling. Through structure and tailoring design, a simple laminate can generally form a monostable structure or a bistable structure. The monostable structure only has one potential energy well, representing the existence of only one stable configuration, as shown in Fig. 1(a). The bistable structure has two potential energy wells, generally, square and rectangular bistable laminates have symmetric and asymmetric local potential energy wells, as shown in Fig. 1 (a) and (b) respectively. The energy difference from the “valley” to the “peak” of the potential energy diagram which is also from the local stable state to the saddle point, is the minimum energy barrier. U_1 and U_2 represent the minimum energy barrier of different configurations. The minimum energy barrier is the minimum energy required to transition from one stable state to the other, and a higher energy barrier means a more stable state.

The advantages of bistable structures have been widely recognized and have been applied in various fields. The theoretical research on such structures regarding stable configuration design, switching path planning, resonance frequency adjustment, and low-energy actuation has laid the foundation for further large-scale application. Some scholars previously pointed out the advantages of dynamic actuation. In comparison to traditional mechanical actuating devices, piezoelectric actuators reduce the weight of the structure while enhancing controllability. The integration of the actuation system and the structure is more conducive to the application of bistable laminates in various fields, which is also the reason for abandoning the vibration exciter. The MFC is adhered to the surface of the laminate. Based on this, this paper further proposes resonant actuation and determines the calculation method for the critical voltage of MFC actuators for resonant actuation.

The present study establishes an integrated framework for modeling, actuation, and optimization of piezoelectrically driven bistable laminates. The main novelties of this work are summarized as follows: (1) A resonant actuation strategy is proposed, which significantly reduces the critical snap-through voltage compared to quasi-static and conventional dynamic actuation. A systematic method for determining the critical voltage and frequency is introduced, addressing a gap in existing dynamic actuation studies. (2) A comprehensive dynamic model is developed using Hamilton’s principle and the Rayleigh-Ritz method, validated against finite element simulations. The model captures multi-mode snap-through behaviors under resonant excitation and enables classification of response types using bifurcation diagrams and Poincaré maps. (3) A multi-objective optimization scheme incorporating Response Surface Methodology (RSM) and the NSGA-II algorithm is implemented to achieve optimal trade-offs between actuation voltage and deformation, with experimental validation of Pareto-optimal designs.

The paper is structured as follows: Section 2 develops finite element and analytical models for bistable laminates with orthogonally bonded MFC actuators. The accuracy of the analytical model is verified by comparing the geometric predictions of the stable configurations against the finite element results, followed by a parametric study on these configurations. Section 3 investigates the static and dynamic actuation strategies and the corresponding structural responses, with a particular focus on delineating the rich nonlinear dynamic phenomena, including snap-through modes, observed during dynamic actuation. Section 4 establishes a surrogate model linking the actuation performance to the laminate geometry, based on which a multi-objective optimization of the dynamic actuation strategy is performed. The optimization results are subsequently validated experimentally. Finally, the main conclusions are summarized in Section 5.

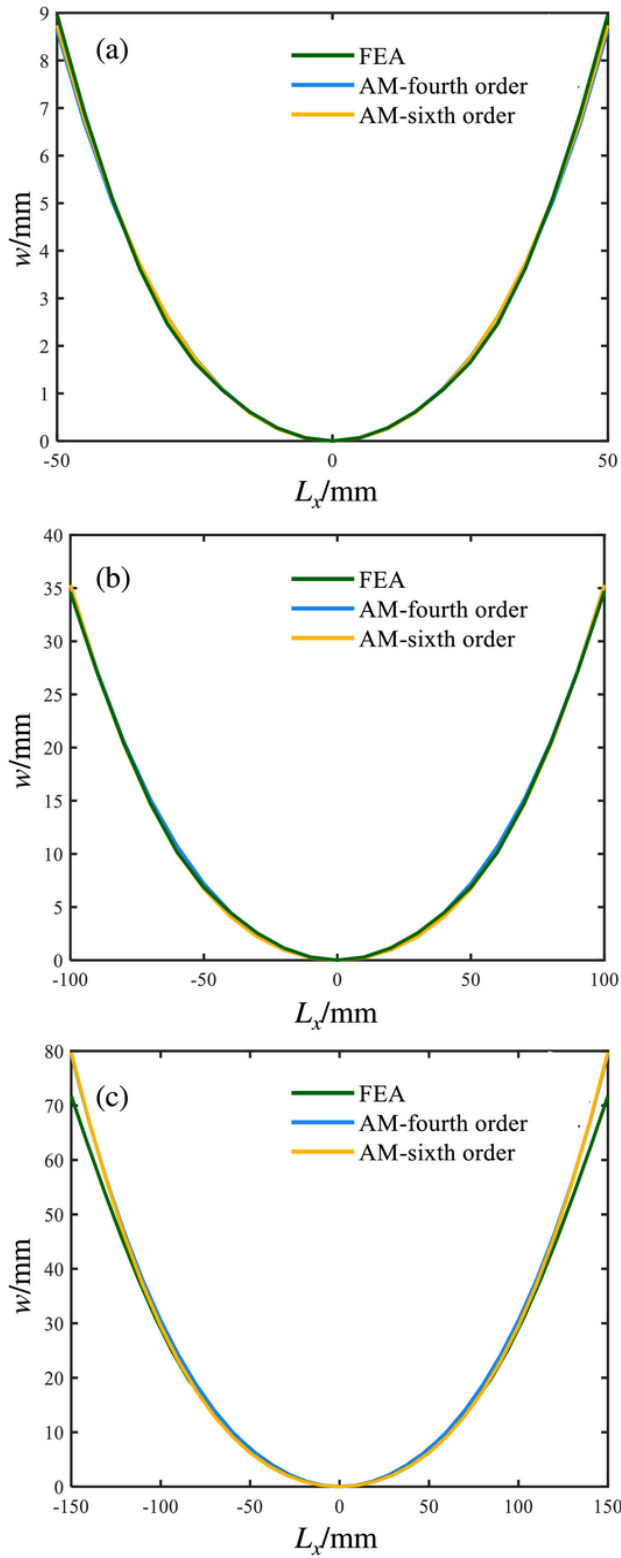


Fig. 5. The cross-section shape of the stable configuration at $y = 0$, (a) $L_x = L_y = 100$ mm, (b) $L_x = L_y = 200$ mm, (c) $L_x = L_y = 300$ mm.

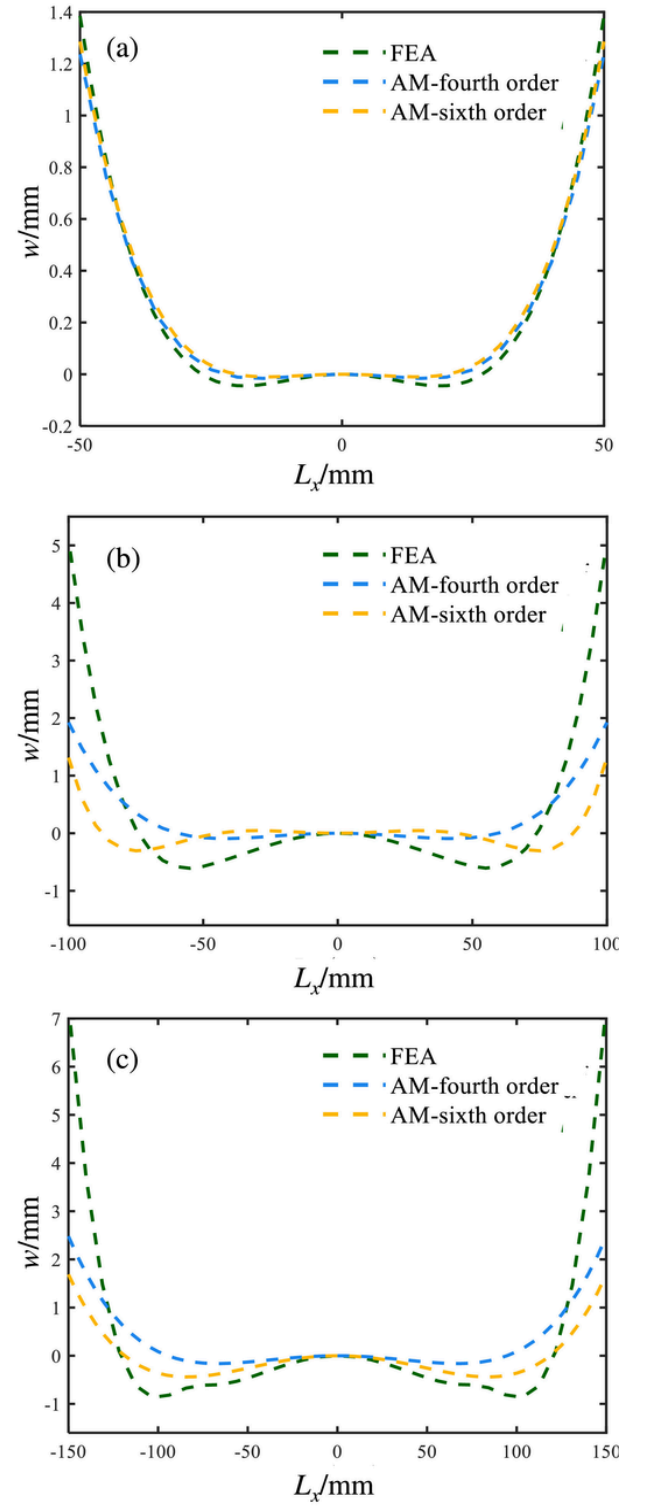


Fig. 6. The cross-section shape of the unstable configuration at $y = 0$, (a) $L_x = L_y = 100$ mm, (b) $L_x = L_y = 200$ mm, (c) $L_x = L_y = 300$ mm.

2. Modelling and parameter analysis of bistable laminates with MFC actuators

This section establishes a dynamic model of an asymmetric orthogonal laminated plate $[0^{\text{MFC}}/0/90/90^{\text{MFC}}]$. Through the inverse piezoelectric effect of the MFC, an externally applied electric field induces the

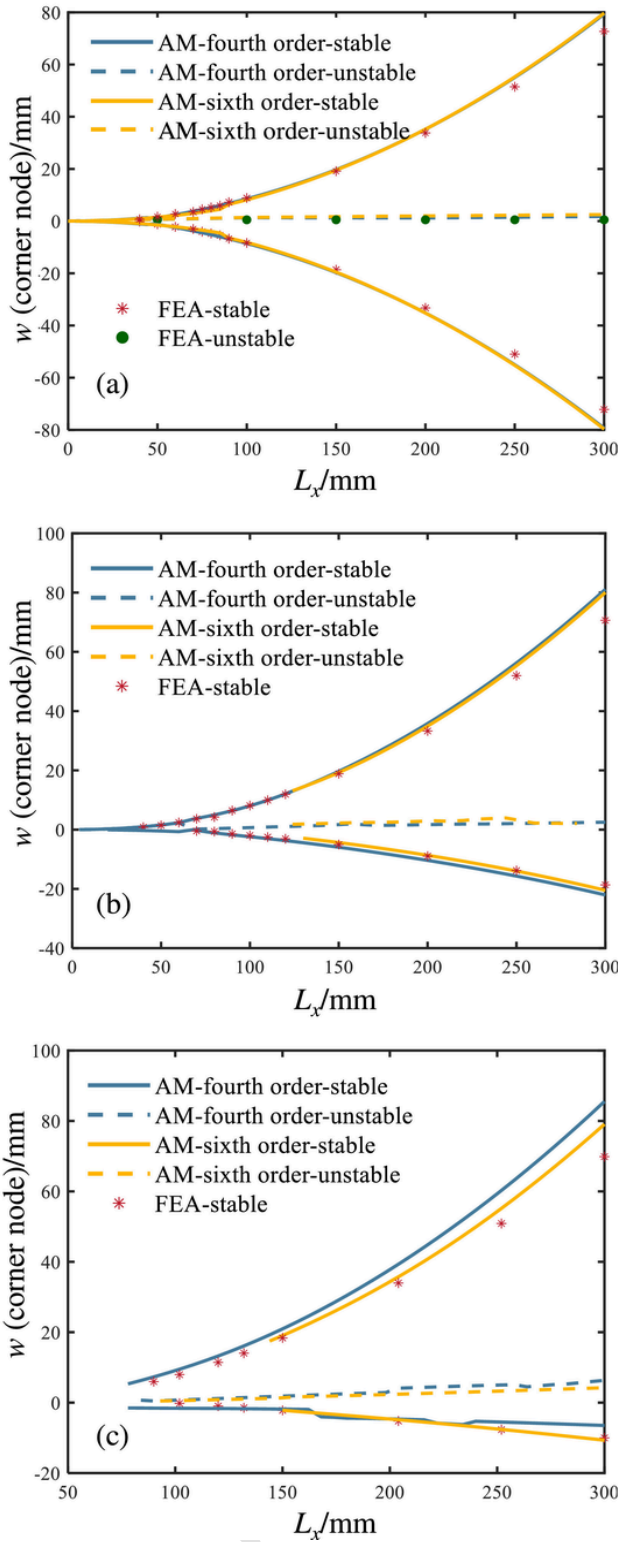


Fig. 7. Parameter analysis of side length of laminates, (a) $AR = 1$, (b) $AR = 2$, (c) $AR = 3$.

material to generate mechanical strain. Since the principal curvatures of the two stable configurations of the structure are orthogonal, the two MFCs are arranged orthogonally in the principal curvature directions of the two configurations to achieve a bidirectional snap-through, as

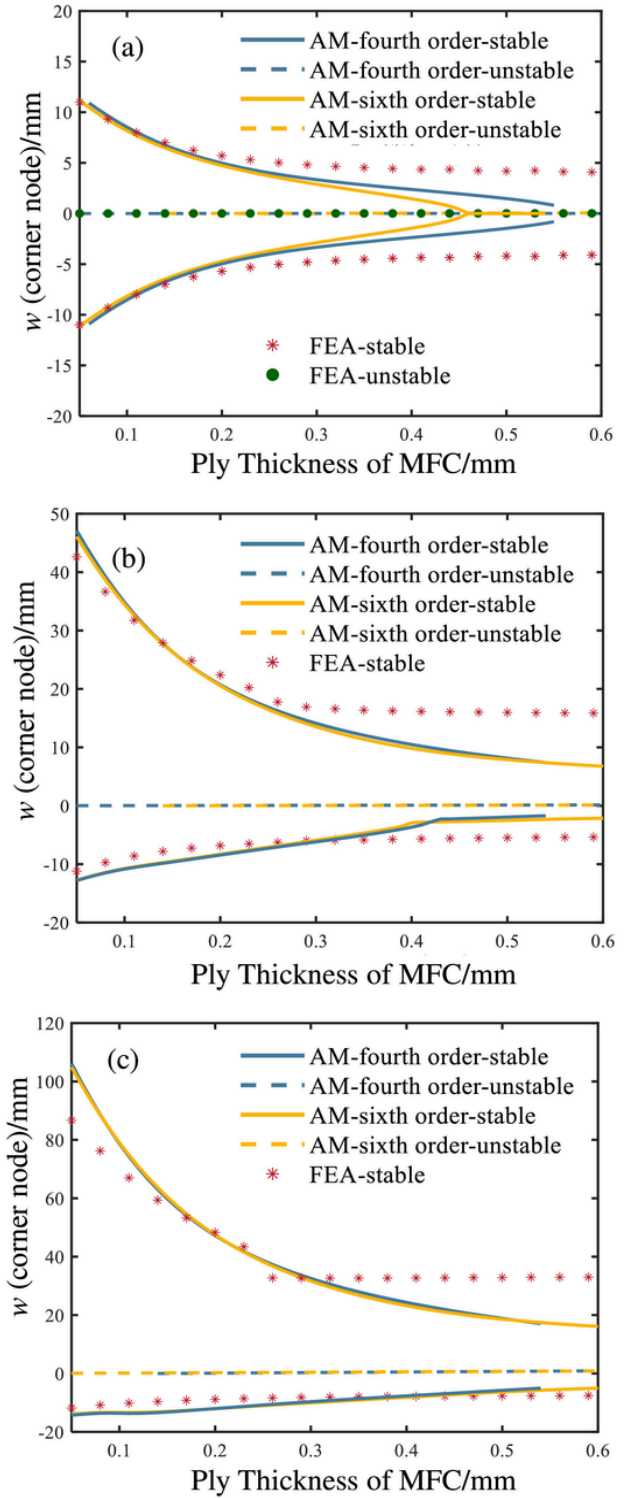


Fig. 8. Parameter analysis of the thickness of actuators, (a) $L_x = 100$ mm, $L_y = 100$ mm, (b) $L_x = 100$ mm, $L_y = 200$ mm, (c) $L_x = 100$ mm, $L_y = 300$ mm.

shown in Fig. 2. To facilitate distinction, MFC1 is used to actuate from stable state A to stable state B, which is positive actuation patterns in this article and MFC2 is used to actuate from stable state B to stable state A, which is negative actuation pattern.

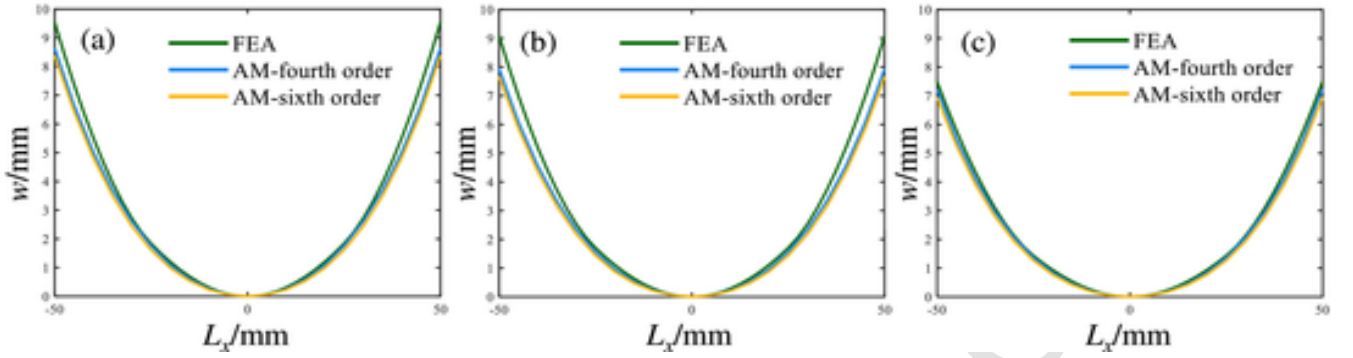


Fig. 9. The cross-section shape of $L_x = 100$ mm, $L_y = 100$ mm laminate at $y = 0$, (a) 0 N, (b) 0.1 N, (c) 0.2 N.

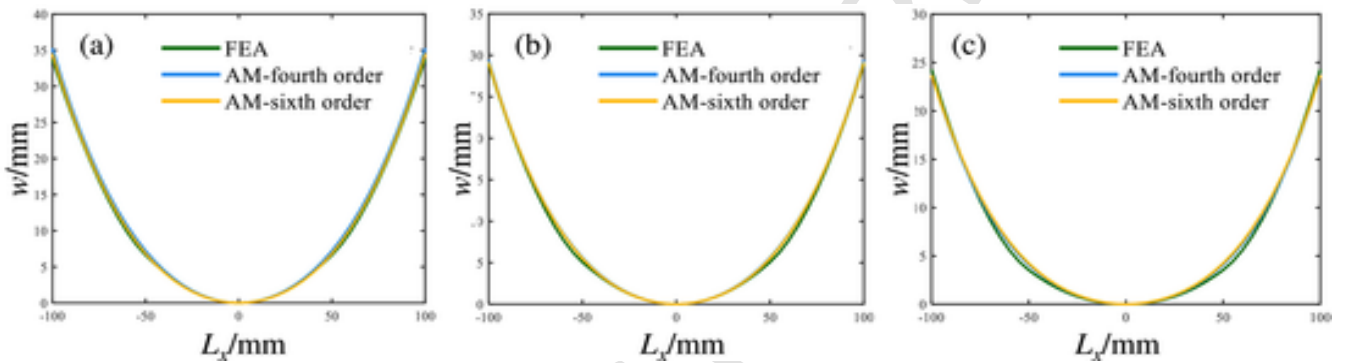


Fig. 10. The cross-section shape of $L_x = 200$ mm, $L_y = 100$ mm laminate at $y = 0$, (a) 0 N, (b) 0.1 N, (c) 0.2 N.

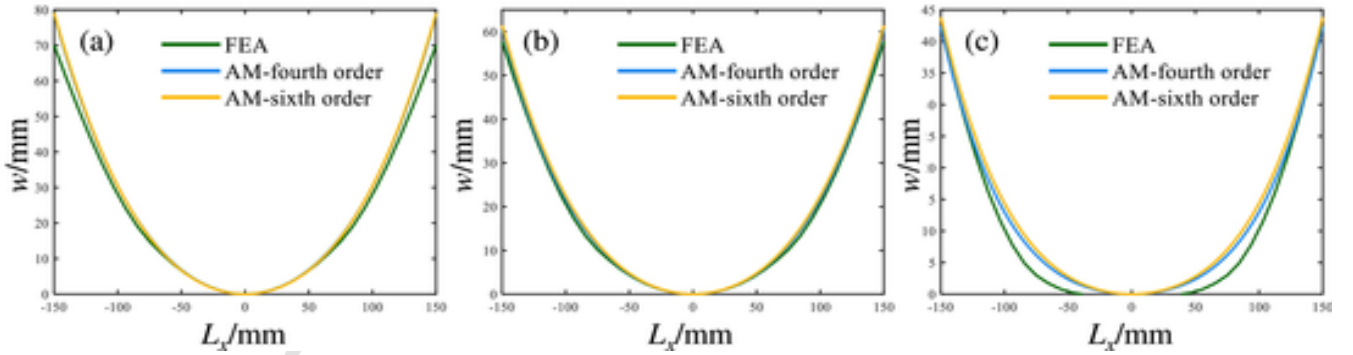


Fig. 11. The cross-section shape of $L_x = 300$ mm, $L_y = 100$ mm laminate at $y = 0$, (a) 0 N, (b) 0.1 N, (c) 0.2 N.

The stable solution of the analytical model can be obtained by constructing the out-of-plane displacement polynomial and solving it. Alternatively, the stable configuration of the bistable laminates can be determined using FEM models. The analytical model has superior computational speed and can give multiple equilibrium solutions, directly yielding both the stable and unstable configurations of bistable laminates. Compared to the analytical models, the advantages of a finite element model lie in that it can predict complex local deformation and provide more detailed information, and it is easier to study bistable laminates with complex boundary conditions and layups. However, its disadvantage is the inability to simultaneously provide stable and unstable configurations of the bistable laminates. It is usually necessary to apply imperfections to the finite element model to make the calculation

result converge to a certain equilibrium configuration, which is time-consuming.

2.1. Finite element model

In this paper, the curing and cooling of the bistable laminate and the adhesion of the MFC are simulated in the commercial finite element software ABAQUS to determine the stresses, strains and shapes during the deformation of the laminates. In order to enable the MFC to adequately control the switching of the two stable configurations, the MFC was arranged orthogonally on the top and bottom of the bistable laminate, as shown in Fig. 2. In this study, a type-P1 d_{33} MFC is adopted, in which the polarization of the piezoelectric material is consistent with

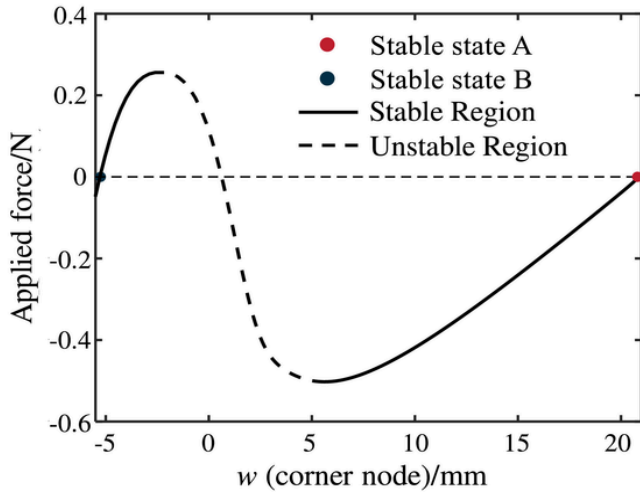


Fig. 12. Force-displacement curve of bistable piezoelectric composite laminates.

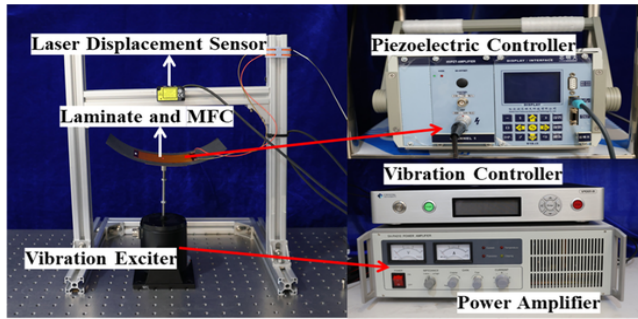


Fig. 13. Experiment equipment.

the direction of the piezoelectric fiber, and it can extend or contract in the direction where is perpendicular to the fiber. In the preparation of specimens, the laminates are made of T700 prepreg material, and the layups are $[0^{MFC}/0/90/90^{MFC}]$. The laid prepreg materials are freely placed in the drying oven. The curing temperature is 200 °C, and it is cooled to room temperature of 20 °C, which is consistent with the simulation calculation. The material properties of laminates are shown in Table 1 [11]. The model of MFC used in the optimization and experiment is M-8557-P1 (CoreMorrow Technology Co., Ltd.). The overall length and width are 103 mm and 64 mm respectively, and the actuation length and width are 85 mm and 57 mm respectively. The thickness of MFC is 0.3 mm.

To be consistent with the basic assumptions of the analytical model, the MFC and the composite laminate are modelled as an integrated shell element in ABAQUS, which MFC is arranged orthogonally on the upper and lower surfaces of the composite laminate. The model is a four-layer shell structure, excluding the “tied” process of MFC and composite materials. The strain imposed by the MFCs is modelled using the thermal expansion analogy [52]. Applying central fixed constraints to the structure, as the finite element model cools from the curing temperature of 200 °C to room temperature of 20 °C, the results converge freely to a cylindrical configuration bent along the long side, referred to as stable state A, as shown in Fig. 3(a). In the absence of defects, for rectangular laminates, the convergence direction is stable state A, while square laminates will converge to an unstable configuration. Restart the analysis step by applying a concentrated force or displacement load in the negative direction of the z axis at the four vertices, which is then re-

leased in the subsequent analysis step. The calculated results can converge to a cylindrical configuration bent along the wide side of the laminate, defined as stable state B, as shown in Fig. 3(b).

In the “Static, General” analysis step, two stable configurations of bistable laminates are obtained by turning on the nonlinear large deformation option of NLGEOM in order to overcome the instability of large deformations in the snap-through process. In the case of piezoelectric bistable laminates, during the calculation of configuration switching using static general analysis, the structure may experience local instabilities that lead to convergence difficulties. In this case, the problem can be transformed into a quasi-static dynamic problem, but it brings additional inertial forces. Local instability can also be controlled by introducing viscous damping by selecting “Specify dissipated energy fraction” of the Automatic stabilization in the analysis step. In this study, the transition from stable state A to stable state B is called forward loading, and vice versa is called reverse loading. At present, the finite element simulation in this paper is unable to simulate the accurate actuation voltage.

Fig. 3 shows the out-of-plane displacement landscapes of the two stable configurations, respectively. It can be seen that MFC mainly affects the displacement of the antisymmetric laminates in the vicinity of the pasted region. The displacement gradually increases from the center to the edge along the direction of the principal curvature, reaching the maximum value at the boundary. The presence of MFC increases the stiffness of the structure and limits the strain and displacement in the central region. Two stable configurations exhibit significantly different in-plane stiffness variations, with stable state B showing an extended state with high in-plane stiffness, whereas stable state A has higher flexibility.

2.2. Analytical model

Single-layer composites are macroscopically homogeneous, linearly elastic, orthotropic and free of initial stress. This section presents a method for characterizing the cured shape and mechanical response of asymmetrically orthogonal bistable laminates. The modeling steps of the analytical model are mainly divided into three steps, cooling the cured composite laminates, integrating MFC into the laminates, and applying voltage to the MFC actuator. A Cartesian coordinate system is established with the center of the laminate as the origin, in which the x axis and y axis are parallel to the length and width of the laminate, respectively. The boundary conditions of the laminate are fixed at the center and free on four sides. The laminate studied in this paper is an elastic plate with a thickness much smaller than the length of its side. According to Kirchhoff hypothesis, the in-plane strain of the laminate is expressed as

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \varepsilon_{xy}^0 \end{bmatrix} + z \begin{bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{bmatrix} \quad (1)$$

where the curvatures are

$$\begin{bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{bmatrix} = \begin{bmatrix} -\frac{\partial^2 w}{\partial x^2} \\ -\frac{\partial^2 w}{\partial y^2} \\ -\frac{\partial^2 w}{\partial x \partial y} \end{bmatrix} \quad (2)$$

The basic assumptions of CLT are the straight-line hypothesis, isonormal hypothesis, plane stress hypothesis, linear elasticity and small deformation hypothesis. The configuration deflection of the bistable laminate is relatively large and does not comply with the linear elasticity and small deformation hypothesis, but only satisfies the other three

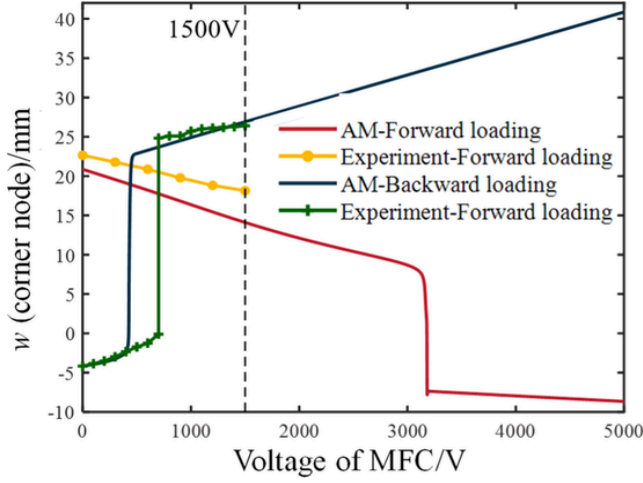


Fig. 14. The displacement curves of quasi-static loading.

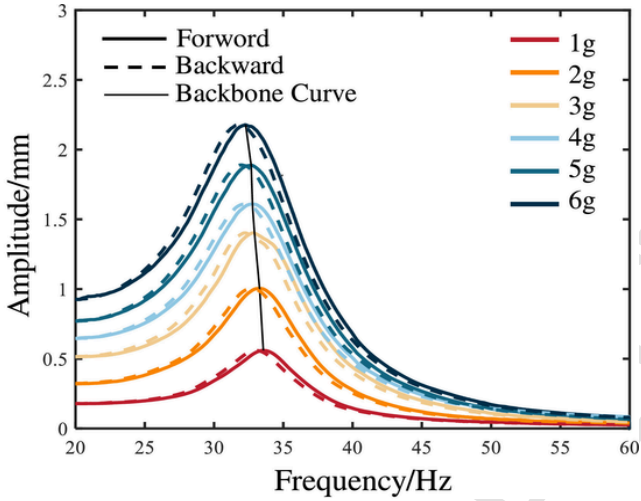


Fig. 15. The backbone curve of bistable piezoelectric laminate.

hypotheses. Considering that the deflection of the bistable plate is larger than the thickness of the laminate, geometric nonlinearity must be taken into account in the geometric equation. According to the von Kármán approximation, the strain-displacement relationship of the bistable laminate is as follows

$$\begin{bmatrix} \epsilon_x^0 \\ \epsilon_y^0 \\ \gamma_{xy}^0 \end{bmatrix} = \begin{bmatrix} \frac{\partial u^0}{\partial x} + \frac{1}{2} \left(\frac{\partial w}{\partial x} \right)^2 \\ \frac{\partial v^0}{\partial y} + \frac{1}{2} \left(\frac{\partial w}{\partial y} \right)^2 \\ \frac{\partial u^0}{\partial y} + \frac{\partial v^0}{\partial x} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \end{bmatrix} \quad (3)$$

Where u^0 , v^0 and w are the mid-plane displacements and out-of-plane displacements respectively. During the curing and cooling process of the laminate, residual thermal stresses are generated without any additional external energy input, so the potential energy of the laminates is

$$\Pi_L = \frac{1}{2} \int_{V_L} \left[(\sigma_x - \sigma_x^T) \epsilon_x + (\sigma_y - \sigma_y^T) \epsilon_y + (\sigma_{xy} - \sigma_{xy}^T) \epsilon_{xy} \right] dV \quad (4)$$

σ and ϵ are the stresses and strains of the laminate, respectively, σ^T and ϵ^T are the free-thermal stresses and strains, which are given by the following formula

$$\begin{cases} \sigma_x = \bar{Q}_{11} \epsilon_x + \bar{Q}_{12} \epsilon_y + \bar{Q}_{16} \gamma_{xy} - \sigma_x^T \\ \sigma_y = \bar{Q}_{12} \epsilon_x + \bar{Q}_{22} \epsilon_y + \bar{Q}_{26} \gamma_{xy} - \sigma_y^T \\ \sigma_{xy} = \bar{Q}_{16} \epsilon_x + \bar{Q}_{26} \epsilon_y + \bar{Q}_{66} \gamma_{xy} - \sigma_{xy}^T \end{cases} \quad (5)$$

$$\begin{cases} \sigma_x^T = \bar{Q}_{11} \epsilon_x^T + \bar{Q}_{12} \epsilon_y^T + \bar{Q}_{16} \gamma_{xy}^T \\ \sigma_y^T = \bar{Q}_{12} \epsilon_x^T + \bar{Q}_{22} \epsilon_y^T + \bar{Q}_{26} \gamma_{xy}^T \\ \sigma_{xy}^T = \bar{Q}_{16} \epsilon_x^T + \bar{Q}_{26} \epsilon_y^T + \bar{Q}_{66} \gamma_{xy}^T \end{cases} \quad (6)$$

Because the coupling stiffness matrix of asymmetric laminates is not zero, there is stretching-bending coupling in the laminate. $\bar{Q}$ is the reduced stiffness term of the laminate, which is calculated according to the following equation

$$\begin{bmatrix} \bar{Q}_{11} \\ \bar{Q}_{12} \\ \bar{Q}_{22} \\ \bar{Q}_{16} \\ \bar{Q}_{26} \\ \bar{Q}_{66} \end{bmatrix} = \begin{bmatrix} C^4 & 2C^2S^2 & S^4 & 4C^2S^2 \\ C^2S^2 & C^4 + S^4 & C^2S^2 & -4C^2S^2 \\ S^4 & 2C^2S^2 & C^4 & 4C^2S^2 \\ C^3S & CS^3 - C^3S & -CS^3 & -2CS(C^2 - S^2) \\ CS^3 & C^3S - CS^3 & -C^3S & 2CS(C^2 - S^2) \\ C^2S^2 & -2C^2S^2 & C^2S^2 & (C^2 - S^2)^2 \end{bmatrix} \begin{bmatrix} Q_{11} \\ Q_{12} \\ Q_{22} \\ Q_{66} \end{bmatrix} \quad (7)$$

where $C = \cos \beta$, $S = \sin \beta$. The relationship between Q and the engineering elastic constants of a single layer is as follows

$$\begin{cases} Q_{11} = \frac{E_1}{1 - \nu_{12}\nu_{21}} \\ Q_{12} = \frac{\nu_{12}E_2}{1 - \nu_{12}\nu_{21}} \\ Q_{22} = \frac{E_2}{1 - \nu_{12}\nu_{21}} \\ Q_{66} = G_{12} \end{cases} \quad (8)$$

The thermal strains depend on the curing temperature

$$\begin{cases} \epsilon_x^T = \alpha_x \Delta T \\ \epsilon_y^T = \alpha_y \Delta T \\ \gamma_{xy}^T = \alpha_{xy} \Delta T \end{cases} \quad (9)$$

ΔT is the temperature difference in the cooling process, where $\alpha_x, \alpha_y, \alpha_{xy}$ are the thermal expansion coefficients, which can be expressed as

$$\begin{cases} \alpha_x = \alpha_1 \cos^2 \beta + \alpha_2 \sin^2 \beta \\ \alpha_y = \alpha_1 \sin^2 \beta + \alpha_2 \cos^2 \beta \\ \alpha_{xy} = (\alpha_1 - \alpha_2) \sin \beta \cos \beta \end{cases} \quad (10)$$

β is the laying angle of the single-layer prepreg board.

Due to the positive piezoelectric effect, the strains ϵ^E are generated inside the polarized ceramic under an applied electric field, and the direction of the electric field determines whether the MFC generates compressive strains or tensile strains. Neglecting the electromechanical coupling inside the MFC, the MFC converts electrical energy into mechani-

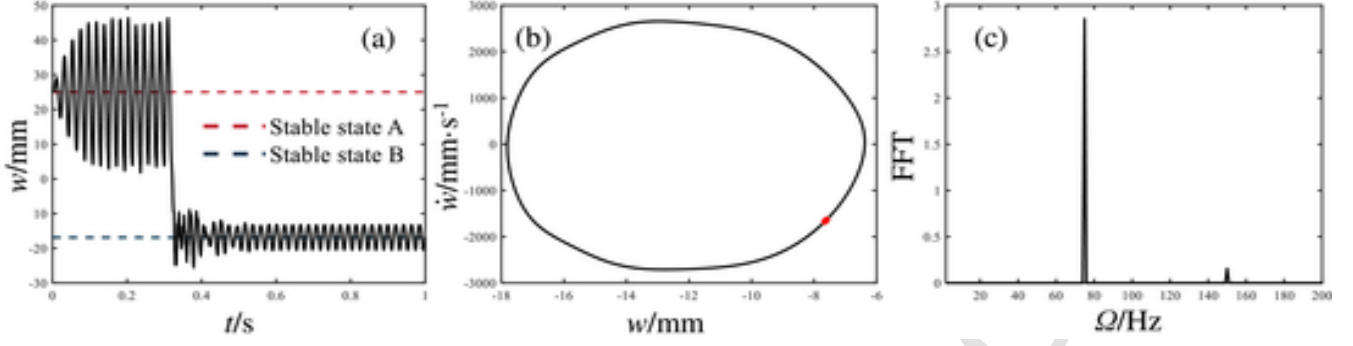


Fig. 16. Transient cross-well oscillation, (a) displacement curve, (b) phase diagram, (c) spectrum map.

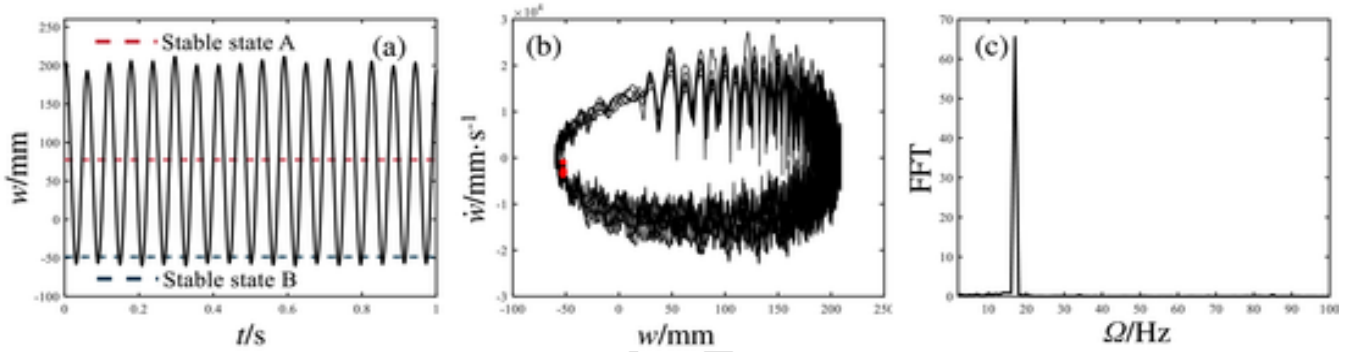


Fig. 17. Continuous cross-well oscillation, (a) displacement curve, (b) phase diagram, (c) spectrum map.

cal energy, so the total potential energy actuated by the MFC is expressed as

$$\Pi_{MFC} = \frac{1}{2} \int_{V_{MFC}} (\sigma_x^e \epsilon_x^e + \sigma_y^e \epsilon_y^e + \sigma_{xy}^e \epsilon_{xy}^e) dV \quad (11)$$

where σ^e and ϵ^e are the stresses and strains of MFC, respectively, given by

$$\begin{cases} \sigma_x^e = \bar{Q}_{11}^e \epsilon_x^e + \bar{Q}_{12}^e \epsilon_y^e + \bar{Q}_{16}^e \gamma_{xy}^e - \sigma_x^E \\ \sigma_y^e = \bar{Q}_{12}^e \epsilon_x^e + \bar{Q}_{22}^e \epsilon_y^e + \bar{Q}_{26}^e \gamma_{xy}^e - \sigma_y^E \\ \sigma_{xy}^e = \bar{Q}_{16}^e \epsilon_x^e + \bar{Q}_{26}^e \epsilon_y^e + \bar{Q}_{66}^e \gamma_{xy}^e - \sigma_{xy}^E \end{cases} \quad (12)$$

σ^E and ϵ^E are piezoelectrically-induced stresses and strains, where the stresses caused by piezoelectric effect are

$$\begin{cases} \sigma_x^E = \bar{Q}_{11}^E \epsilon_x^E + \bar{Q}_{12}^E \epsilon_y^E + \bar{Q}_{16}^E \gamma_{xy}^E \\ \sigma_y^E = \bar{Q}_{12}^E \epsilon_x^E + \bar{Q}_{22}^E \epsilon_y^E + \bar{Q}_{26}^E \gamma_{xy}^E \\ \sigma_{xy}^E = \bar{Q}_{16}^E \epsilon_x^E + \bar{Q}_{26}^E \epsilon_y^E + \bar{Q}_{66}^E \gamma_{xy}^E \end{cases} \quad (13)$$

and the strains ϵ^E are given by

$$\begin{bmatrix} \epsilon_x^E \\ \epsilon_y^E \\ \epsilon_{xy}^E \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \bar{d}_{33} & 0 & 0 \\ \bar{d}_{32} & 0 & 0 \\ \bar{d}_{31} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \bar{d}_{15} \\ 0 & \bar{d}_{24} & 0 \end{bmatrix} \begin{bmatrix} U \\ \Delta t \\ 0 \end{bmatrix} \quad (14)$$

where U is the electric field strength, Δt is the distance between the two electrodes, and d is the piezoelectric coefficient. Since the thickness of the MFC is only 0.3 mm, the stresses and strains in the z direction of the plate thickness are ignored. The piezoelectric matrix requires coordinate transformation

$$\bar{d} = d \cdot T \quad (15)$$

Transformation matrix T is

$$T = \begin{bmatrix} C^2 & S^2 & 0 & CS & 0 & 0 \\ S^2 & C^2 & 0 & -CS & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -2CS & 2CS & 0 & C^2 - S^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & C & -S \\ 0 & 0 & 0 & 0 & S & C \end{bmatrix} \quad (16)$$

Depending on the transformation of rectangular coordinates in space, stress and strain can be flexibly converted between the principal coordinate system of the material and the spatial global coordinate system.

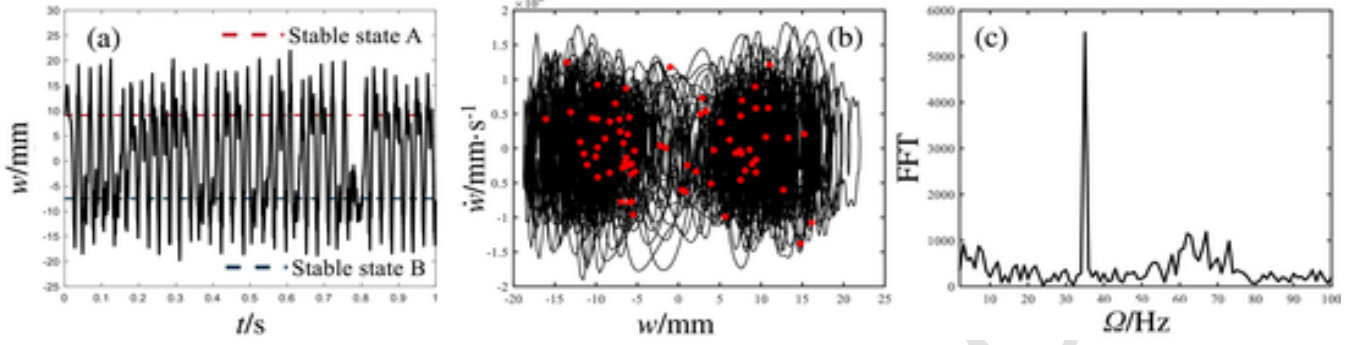


Fig. 18. Chaotic oscillation, (a) displacement curve, (b) phase diagram, (c) spectrum map.

According to Hamilton's principle, the time integral of the total virtual work is zero.

$$\delta \int_{t_0}^{t_1} (T - \Pi) dt + \delta \int_{t_0}^{t_1} W dt = 0 \quad (17)$$

where T , Π and W are the kinetic energy, strain energy and external work of the structure respectively, and the total potential energy is the sum of the potential energy of the composite laminates and MFC:

$$\Pi = \Pi_L + \Pi_{MFC} \quad (18)$$

Kinetic energy can be expressed as

$$T = T_L + T_{MFC} = \frac{1}{2} \int_{V_L} \rho_L (\dot{u}^2 + \dot{v}^2 + \dot{w}^2) dV + \frac{1}{2} \int_{V_{MFC}} \rho_{MFC} (\dot{u}^2 + \dot{v}^2 + \dot{w}^2) dV \quad (19)$$

The static loading in this paper is applied via four concentrated forces at the four corners of the plate $x = \pm \frac{L_x}{2}$ and $y = \pm \frac{L_y}{2}$, where L_x and L_y are the length and width of laminate, and the external work is

$$W = 4Fw \left(\frac{L_x}{2}, \frac{L_y}{2}, t \right) - \int_{-\frac{L_x}{2}}^{\frac{L_x}{2}} \int_{-\frac{L_y}{2}}^{\frac{L_y}{2}} C \frac{\partial w}{\partial t} dx dy \quad (20)$$

In the formula, the first part is the work done by the external force and the second part is the work done by the damping force. C is the damping coefficient, and for ease of calculation, Rayleigh damping is used to express the damping as a linear equation with respect to the mass matrix M and the stiffness matrix K

$$C = \alpha M + \beta K \quad (21)$$

where α and β are the mass proportionality coefficient and stiffness proportionality coefficient, respectively. Since bistable vibrations tend to occur at low frequencies, the role of stiffness damping is neglected, and β is set to zero.

Since the actual strain distribution of the mid-plane cannot be obtained, the approximate solution is obtained using the Rayleigh-Ritz method. The results show that the cooled laminate is not perfectly cylindrical and the bending curvature is not constant. Taking into account the boundary conditions and symmetry of the laminate, it is assumed that the out-of-plane displacement function of the laminate

$$w = a_0(t)x^2 + a_1(t)y^2 + a_2(t)x^4 + a_3(t)y^4 + a_4(t)x^2y^2 + a_5(t)x^6 + a_6(t)y^6 \quad (22)$$

Compared with the assumption of the mid-plane displacement field, the strain field can simplify the calculation. The following set of polynomials is used to approximate the mid-plane strain

$$\begin{cases} \varepsilon_x^0(t) = b_0(t) + b_1(t)y^2 + b_2(t)y^4 + b_3(t)x^2y^2 + b_4(t)y^6 \\ \varepsilon_y^0(t) = c_0(t) + c_1(t)x^2 + c_2(t)x^4 + c_3(t)x^2y^2 + c_4(t)x^6 \end{cases} \quad (23)$$

In order to verify the accuracy of the model and improve the computational efficiency, a fourth-order model is proposed. The accuracy of the two analytical models and the finite element model are compared in the following section.

$$w = a_0(t)x^2 + a_1(t)y^2 + a_2(t)x^4 + a_3(t)y^4 + a_4(t)x^2y^2 \quad (24)$$

$$\begin{cases} \varepsilon_x^0(t) = b_0(t) + b_1(t)y^2 + b_2(t)y^4 + b_3(t)x^2y^2 \\ \varepsilon_y^0(t) = c_0(t) + c_1(t)x^2 + c_2(t)x^4 + c_3(t)x^2y^2 \end{cases} \quad (25)$$

The unknown coefficients $a_i(t)$, $b_i(t)$ and $c_i(t)$ are generalized coordinates in the model, and the mid-plane displacements $u^0(t)$ and $v^0(t)$ can be obtained by integration

$$\begin{cases} u^0(t) = \int_{-\frac{L_x}{2}}^{\frac{L_x}{2}} \left(\varepsilon_x^0 - \frac{1}{2} \left(\frac{\partial w(t)}{\partial x} \right)^2 \right) dx \\ v^0(t) = \int_{-\frac{L_y}{2}}^{\frac{L_y}{2}} \left(\varepsilon_y^0 - \frac{1}{2} \left(\frac{\partial w(t)}{\partial y} \right)^2 \right) dy \end{cases} \quad (26)$$

Based on Hamilton's principle, 17 and 13 nonlinear differential equations about generalized coordinates and time are derived. The nonlinear equations are solved using the "NDSolve" function in MATHEMATICA, which saves more time compared to the "Implicit, Dynamic" analysis step in finite element simulations. Multiple equilibrium solutions can be obtained, corresponding to stable and unstable configurations respectively. The stability of the solution is determined by judging whether the Jacobian matrix is positive definite. When there are non-positive eigenvalues of the Jacobian matrix, the matrix is not positive definite and the structure is unstable. The solutions of the following system of equations represent the obtained configuration

$$\begin{cases} f_i = \frac{\partial \Pi}{\partial a_i} = 0 \\ g_i = \frac{\partial \Pi}{\partial b_i} = 0 \\ p_i = \frac{\partial \Pi}{\partial c_i} = 0 \end{cases} \quad (27)$$

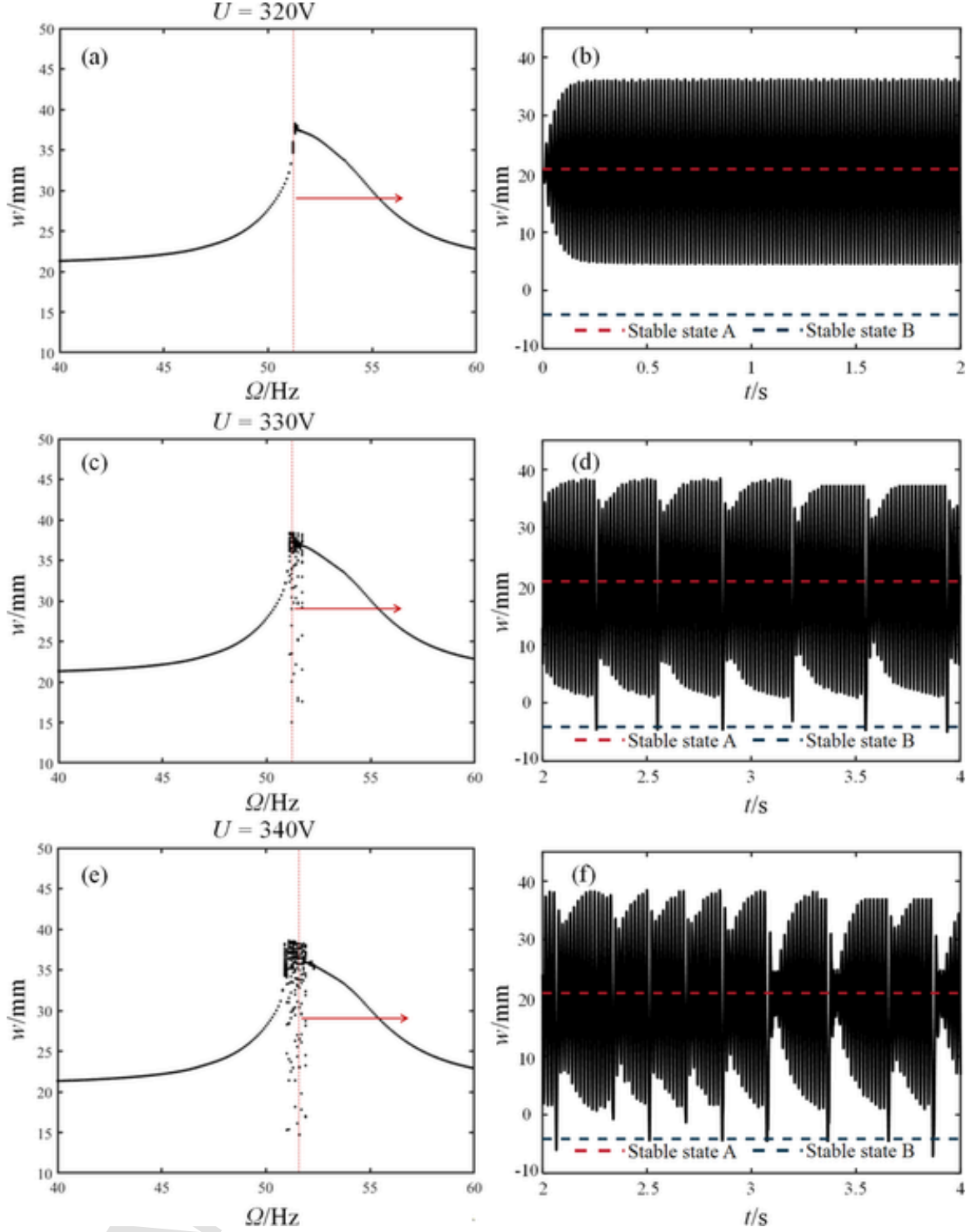


Fig. 19. Poincaré map and displacement time history of stable state A at different excitation frequencies, (a) $U = 320$ V, Poincaré map, (b) $U = 320$ V, $\Omega = 51.2$ Hz, displacement curve, (c) $U = 330$ V, Poincaré map, (d) $U = 330$ V, $\Omega = 51.2$ Hz, displacement curve, (e) $U = 340$ V, Poincaré map, (f) $U = 340$ V, $\Omega = 51.2$ Hz, displacement curve.

Substituting the obtained solutions into the following formula to find the Jacobian matrix

$$J = \frac{\partial (f_i, g_i, p_i)}{\partial (a_i, b_i, c_i)} \quad (28)$$

Consistent with the specimens and the finite element simulation, the midpoint of the laminated plate is fixed. The curing temperature is 200°C , then cooled to room temperature of 20°C . By changing the trial function set, the model can converge to stable or unstable configurations, as shown in Fig. 4.

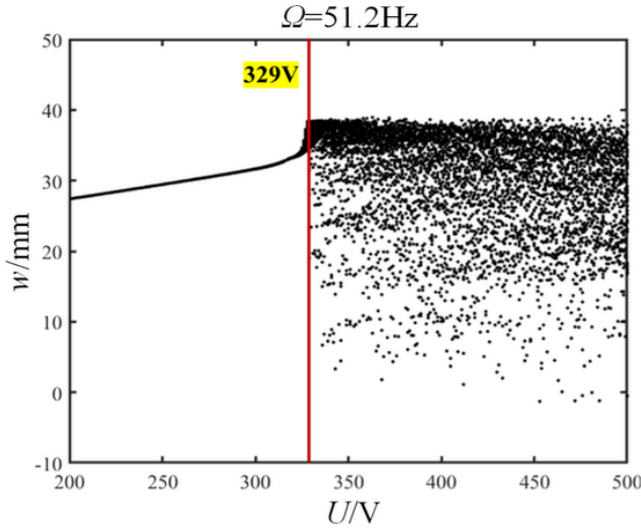


Fig. 20. Bifurcation diagram of stable state A.

2.3. Verification of model accuracy

The bistable laminates with side lengths of 100 mm, 200 mm, and 300 mm were analyzed using three approaches: the finite element analysis (FEA), the fourth-order analytical model (AM-fourth order), and the sixth-order analytical model (AM-sixth order), as detailed in Sections 2.1 and 2.2. In the subsequent configuration prediction and parameter analysis, the edge length of MFC in all structures is half of the corresponding edge length of the laminate, and the thickness is 0.1 mm. Then, the cross-sectional shapes at $y = 0$ of the stable configurations and the unstable configurations are compared with each other. In the following configuration prediction and parameter analysis, the side length of the MFC is half of the side length of the laminate. Fig. 5 shows the cross-section shape of the stable configuration parallel to the directrix of the cylinder, where green, yellow and blue lines correspond to the prediction results of the finite element, the fourth-order analytical model and the sixth-order analytical model, respectively.

For the stable configuration, the cross-sectional shape is parabolic, and the deflection of the vertex increases with the increase of the side length of the laminate. The errors of the analytical model and the finite element model mainly appear at the boundaries of the composite laminate and the MFC, and increase with the increase of the side length. When the side length is 100 mm, the sixth-order model is closer to the finite element results at the edge of the laminate. When the side length is 200 mm, the prediction results of the fourth-order model and the sixth-order model are almost the same at the boundary of the laminate, while the prediction results given by the sixth-order model are very close to the finite element method at the boundary of the MFC. When the side length is 300 mm, the errors at the boundary of both laminates and MFC are further increased. Except for the boundary, the results predicted by the theoretical model and the simulation model are almost identical. In finite element analysis, due to the increase of degrees of freedom, the free-edge effect of the structure can be better predicted. This phenomenon can be demonstrated by the thick plate theory, while the CLT is not capable to predict this phenomenon. This is due to the differences in the expression of torque and transverse shear force at the free boundary between the two.

In the finite element model (simulated in ABAQUS), square laminates without any defects converge to an unstable configuration in the absence of external loads. The cross-sectional shape of the unstable configuration of a square orthogonal laminate is not a parabola, but close to a W-shape. This configuration exhibits consistent bending curvature

and cross-sectional shape along the x and y axes, as shown in Fig. 6. As the side length of the structure increases, the concavity and convexity of the W-shape become more pronounced, and discrepancies between the models gradually increase. Compared to the fourth-order model, the shape predicted by the sixth-order model aligns more closely with the results given by the finite element model. The computational efficiency of the analytical model is much higher than that of the finite element model while ensuring the computational accuracy. The prediction accuracy of the fourth-order model is sufficient, while the sixth-order model is closer to the calculation results of the finite element model at the boundary.

2.4. Parameter analysis

Determining whether a laminate exhibits bistability is crucial for bistable structural design and optimization, while manufacturing methods and material structure parameters are being the primary influencing factors. The analytical model can accurately predict the stable configurations of asymmetric composite laminates, but the prediction of bifurcation points of bistable laminates needs further exploration. Taking the length L_x , width L_y , Aspect ratio (AR) and thickness of the adhesive layer as the design variables, geometrical parameter analysis is carried out to investigate the dimensional range of bistability and bifurcation phenomenon of piezoelectric asymmetric composite laminate. The bifurcation diagrams of the vertex deflections at each edge length are depicted for $AR = 1, 2$ and 3 , respectively. The vertex deflections of the different laminates are shown in Fig. 8, and the horizontal coordinate is the length of the laminate, L_x . The blue and yellow lines are the prediction results of the fourth-order model and the sixth-order model, respectively, and the solid and dashed lines represent stable and unstable configurations. The red and green dots represent stable and unstable configurations predicted by the finite element model, respectively. The calculation results show that there are some errors in the prediction results in both the analytical model and the finite element model.

As shown in Fig. 7(a), for a square asymmetric laminate with an aspect ratio of 1, the predictions of both the finite element and analytical models gradually increase as the side length increases. The three prediction results match well with each other, while the finite element results are relatively small in the large deflection area. As shown in Fig. 7 (b) and (c), for rectangular laminates with aspect ratios of 2 and 3, the sixth-order model cannot converge to a stable configuration when the side length is small, so the existence of bifurcation points cannot be predicted accurately. When the side length is larger, the results of the sixth-order model align more closely with those obtained from finite element analysis. The fourth-order model can accurately predict the out-of-plane displacement of orthogonal laminates with different sizes and demonstrates good convergence in the small-size areas. However, its calculation errors may occur in larger-size areas. The critical size of the laminates that possess bistability increases with the increase of the aspect ratio.

Fig. 8 depicts the bifurcation diagram of a bistable laminate with different side lengths as $L_x = 100$ mm, 200 mm, 300 mm, $L_y = 100$ mm, and the thickness of MFC and the adhesive layer gradually increases from 0.05 mm to 0.6 mm. For bistable laminates in the analytical model, as the thickness increases, the deformation of the structure decreases, eventually leading to the loss of bistability. In the finite element model, the thickness gradually loses its effect on the deformation after increasing to a certain value. When the thickness is < 0.3 mm, the prediction results of the analytical model, especially the sixth-order model, almost agree with the simulation model. For rectangular bistable laminates, the MFC thickness has a much greater effect on the stable state A than on the stable state B. Similarly, when the plate thickness is small, the predictions of the three models are more consistent. However, when the MFC thickness exceeds a certain value, the finite el-

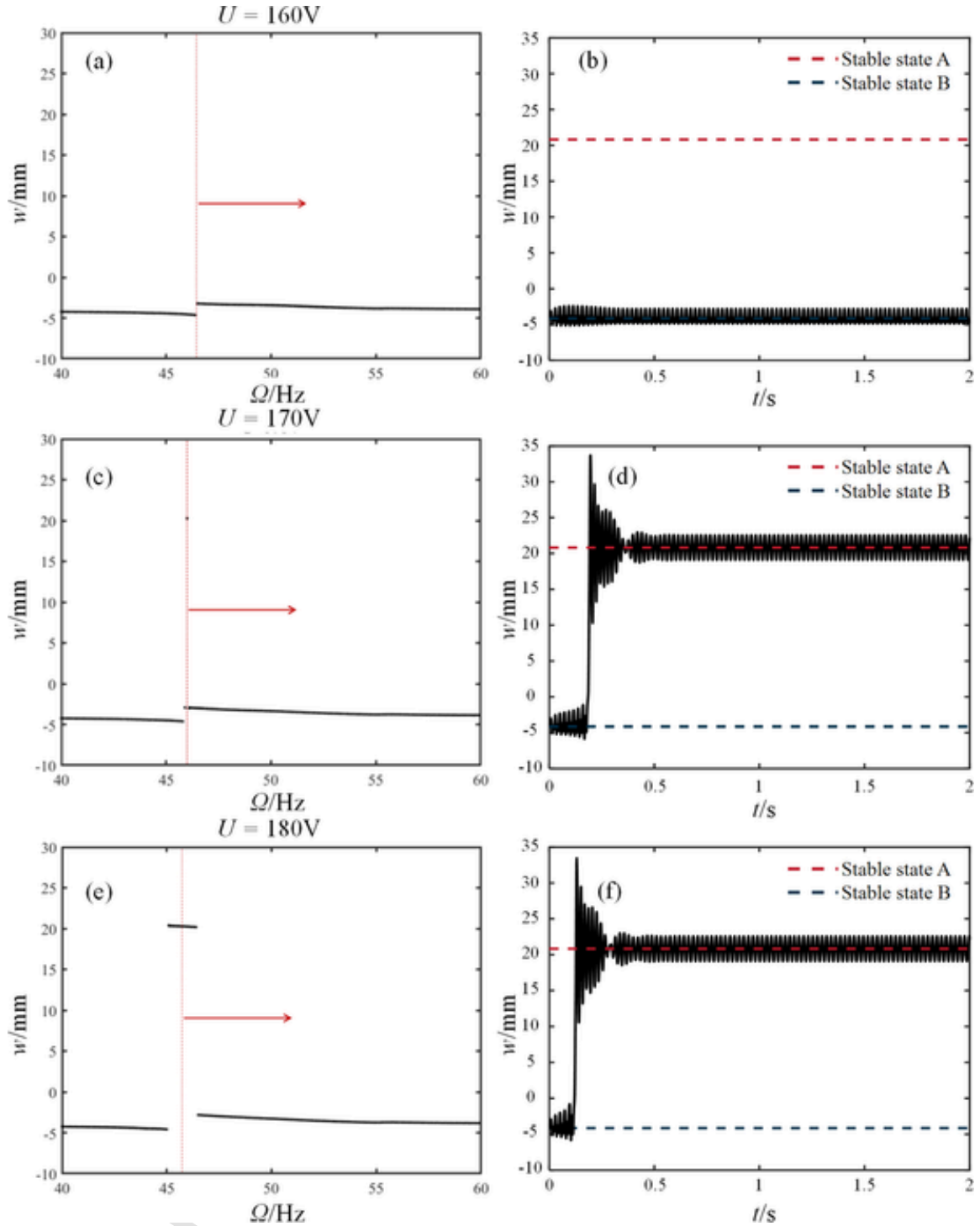


Fig. 21. Poincaré map and displacement time history of stable state A at different excitation amplitudes, (a) $U = 160V$, Poincaré map, (b) $U = 160V$, $\Omega = 46.1$ Hz, displacement curve, (c) $U = 170V$, Poincaré map, (d) $U = 170V$, $\Omega = 46.1$ Hz, displacement curve, (e) $U = 180V$, Poincaré map, (f) $U = 180V$, $\Omega = 46.1$ Hz, displacement curve.

ement model and the analytical model show distinct different responses as the thickness changes.

After the above analysis and settings, the prediction results of the analytical models and the finite element simulation model are consistent in most cases. Increasing the order or degrees of freedom of the analytical model does not lead to a significant improvement of prediction accuracy. The sixth-order model aligns better with the finite element re-

sults in the large-scale area, while the fourth-order model converges more easily in the small-scale area. Since engineering applications are less likely to involve smaller bistable laminates or thicker MFCs, the sixth order model is chosen for subsequent calculations, due to its balance of accuracy and efficiency in computation.

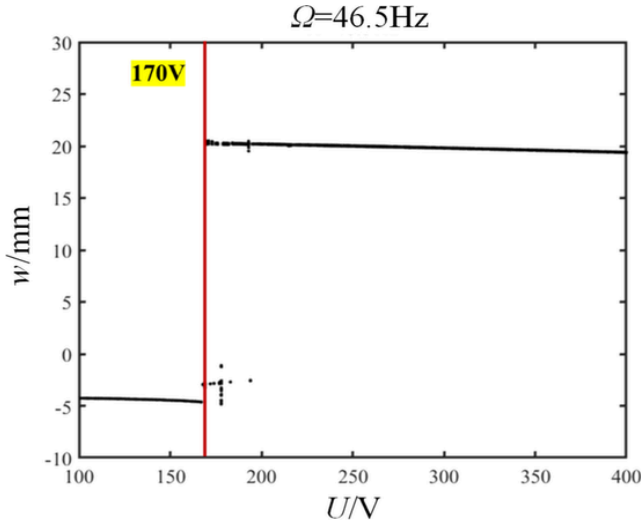


Fig. 22. Bifurcation diagram of stable state B.

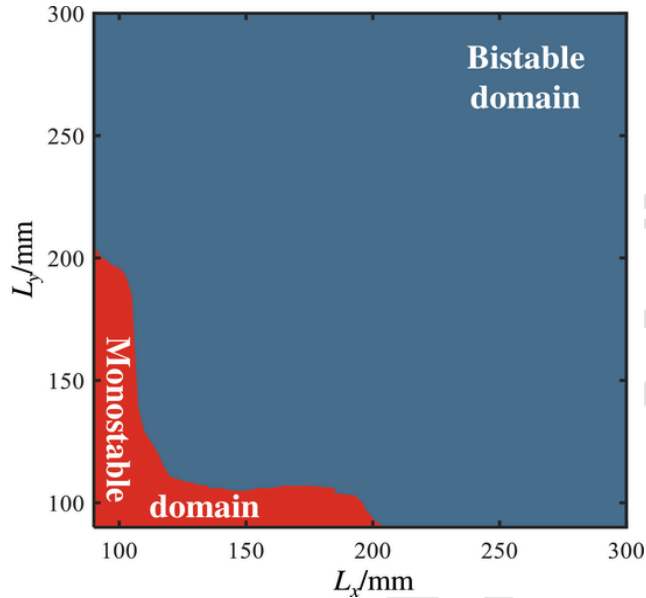


Fig. 23. Bistable domain and monostable domain of asymmetric composite laminates.

3. Mechanical analysis and actuation of bistable laminate

During deformation, the flexible structure achieves the desired response through elastic deformation, so in addition to external loads, the actuation must counter the inherent structural stiffness of the system. Both the previous research and the research in this paper show that static actuation makes it difficult to achieve snap-through of laminates with large deflection. The resonant actuation amplifies the maximum displacement of the structure and ensures that the inertia and stiffness forces cancel each other out during the desired actuation cycle, thus minimizing the actuation requirements.

3.1. Static analysis of bistable laminate

After obtaining the stable configuration of the piezoelectric bistable laminate, the accuracy of the static model is further verified. The length

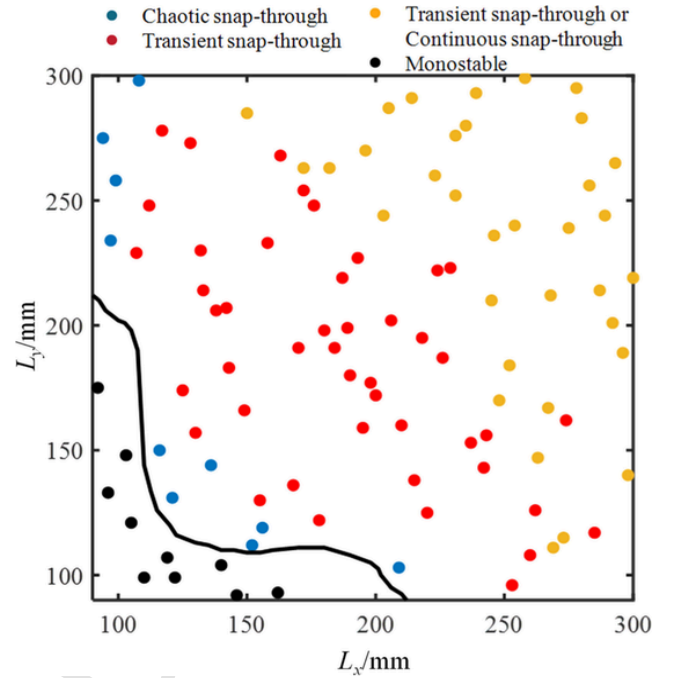


Fig. 24. The snap-through modes of training set.

L_x is set as 100 mm, 200 mm, and 300 mm, respectively, the width L_y is 100 mm, and the layup was unchanged as $[0^{MFC}/0/90/90^{MFC}]$. The midpoint of the laminate remains fixed. Concentrated forces of 0 N, 0.1 N, and 0.2 N are applied to each of the four vertices of the laminate, respectively. Figs. 9–11 compare the cross-section shapes of the stable configurations predicted by the sixth-order analytic model, the fourth-order analytic model and the finite element model at $y = 0$, respectively. For a square laminate with a side length of 100 mm, as shown in Fig. 9, the results of the analytical model are smaller than the predictions of the finite element model when a concentrated force of 0 N and 0.1 N is applied, with the error being largest at the boundary. When the concentrated force is 0.2 N, the prediction results of the three models have very small errors. For laminates with a size of 200 mm $\times$ 100 mm, as shown in Fig. 10, the analytical and simulation results are relatively close to each other, and the errors mainly appear at the edges of the MFC. For the 300 mm $\times$ 100 mm laminate, as shown in Fig. 11, when no force is applied, the deflection of the vertices predicted by the analytical model is approximately 10 mm greater than that predicted by the finite element model. When the concentration force is 0.1 N, the cross sections of the three methods are almost the same. When the concentration force is 0.2 N, the error of the structure is greatest at the periphery of the MFC, while the prediction results are similar at the maximum deflection.

The force-displacement curve of the piezoelectric bistable composite laminate is consistent with that of ordinary bistable composite laminate, which indicates that the snap-through form is the same under a quasi-static loading. As shown in Fig. 12, The solid and dashed lines in the figure represent segments with the positive and negative stiffness, corresponding to the stable and unstable states of the bistable laminate, respectively. The intersection of the solid line with $y = 0$ represents the stable configuration of the structure, and the intersection of the dashed line with $y = 0$ is the unstable equilibrium point. The positive stiffness segment is considered to be the passive deformation caused by absorbing external energy, while the negative stiffness segment is the autonomous deformation caused by the release of internal potential energy. As shown in Fig. 12, that the critical load of transitioning from stable state A to stable state B is about -0.502 N, while the reverse snap-

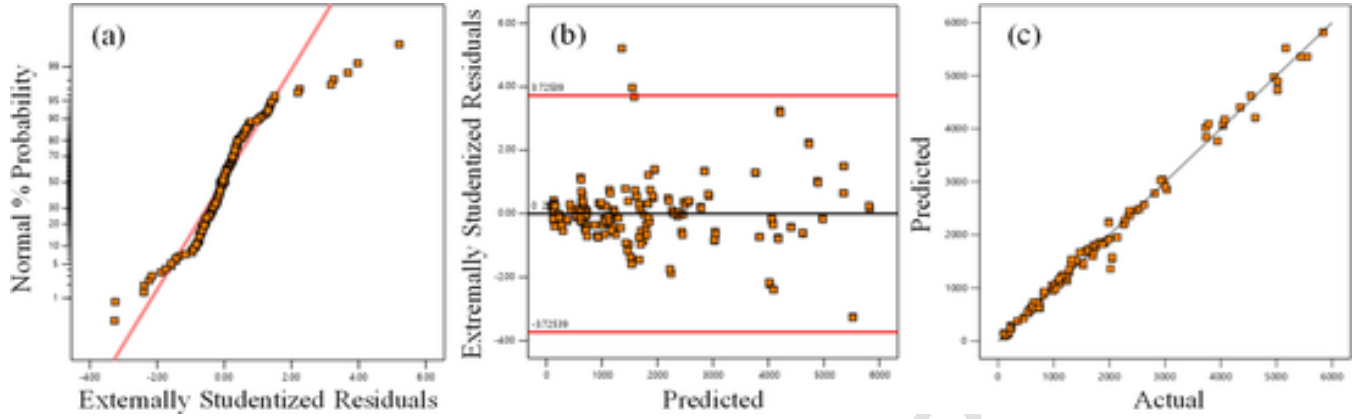


Fig. 25. Critical voltage comparison between the analytical model and surrogate model, (a) distribution of residuals, (b) distribution of residuals and predicted values, (c) distribution of predicted values and actual values.

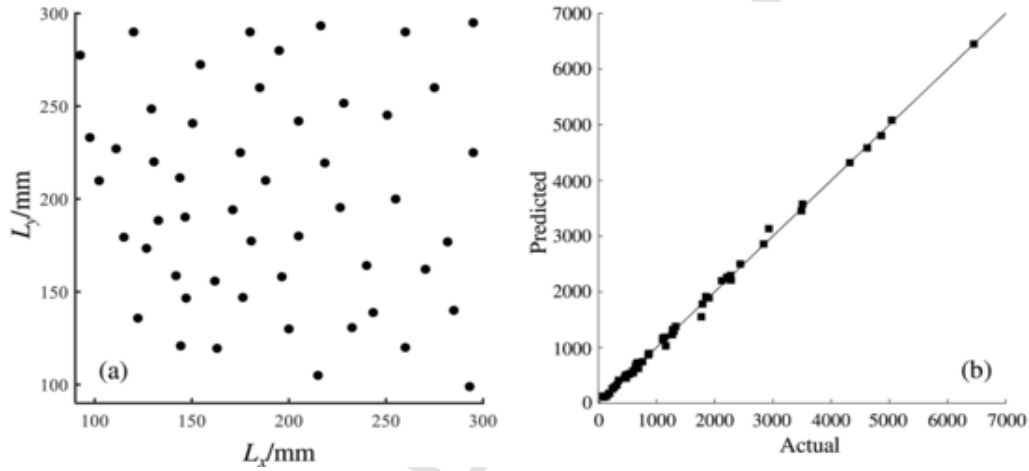


Fig. 26. Model verification, (a) parameter distribution of 50 verification points, (b) distribution of predicted critical voltage values and actual critical voltage values.

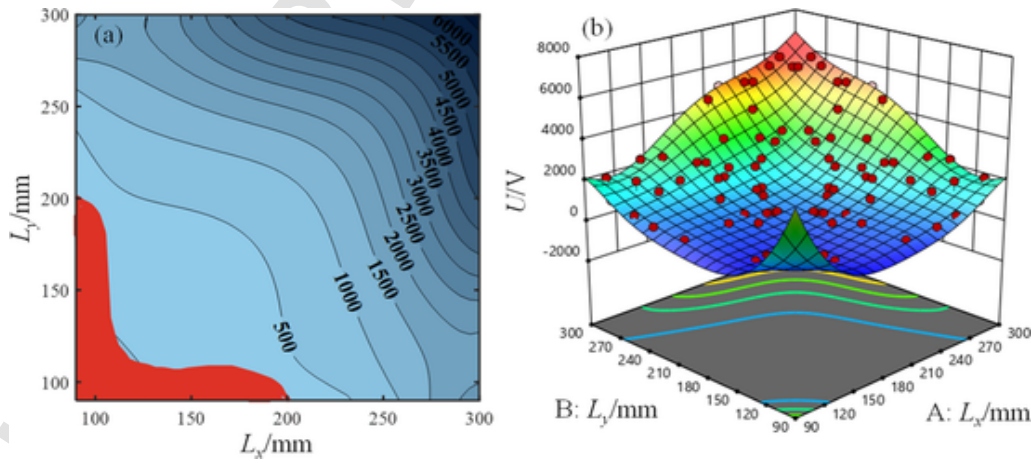


Fig. 27. Critical voltage amplitudes of MFC-actuated bistable laminates, (a) two-dimensional contour plot, (b) three-dimensional plot.

through load is around 0.255 N, demonstrating that the transition from stable state B to stable state A is more easy. Before the external load reaches the critical value, when the external force is withdrawn, the structure will return to its original configuration. After the external

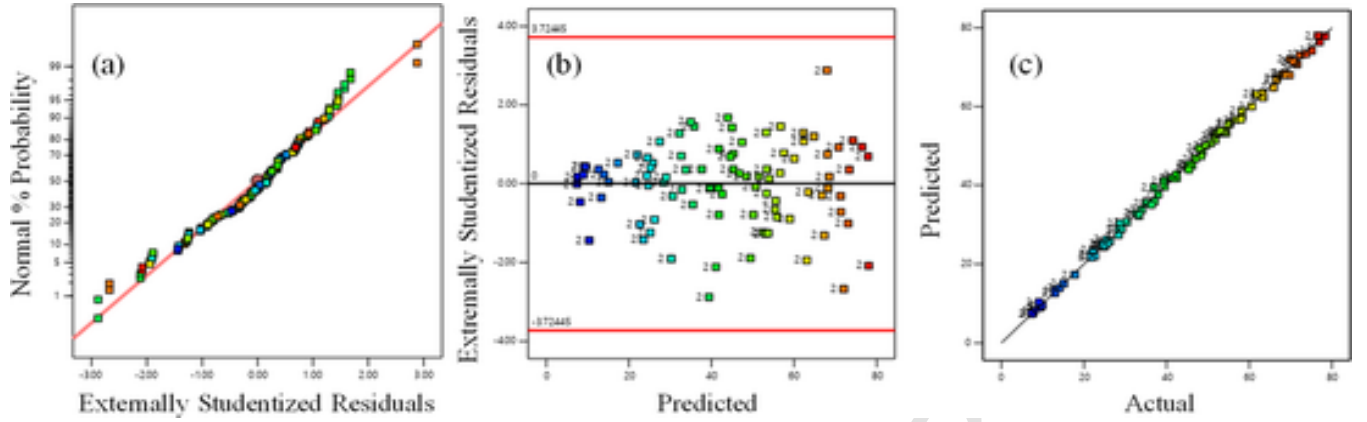


Fig. 28. Vertex deflections comparison between the analytical model and surrogate model, (a) distribution of residuals, (b) distribution of residuals and predicted values, (c) distribution of predicted vertex deflections values and actual vertex deflections values.

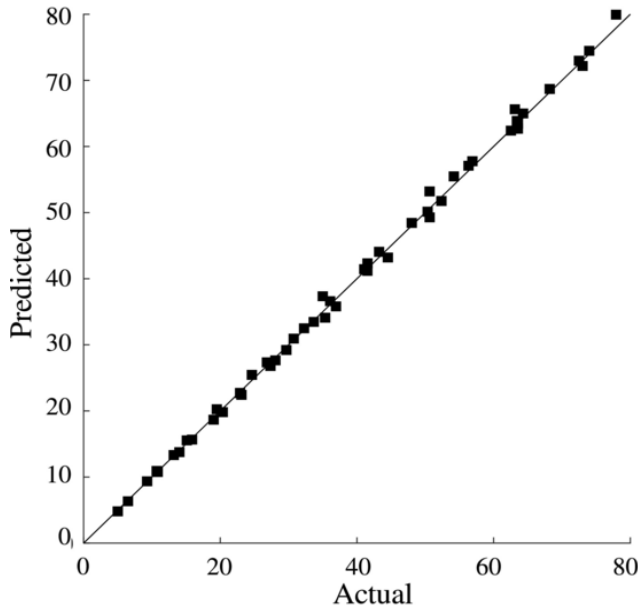


Fig. 29. Distribution of predicted vertex deflections values and actual vertex deflections values in the further verification.

load reaches a critical value, removing the external force will reach the other stable state.

3.2. The quasi-static actuation

In the previous chapter, precise stable configurations and force-displacement load curves of bistable piezoelectric laminates at zero voltage have been obtained. Quasi-static actuation is carried out on $200 \text{ mm} \times 100 \text{ mm}$, $[0^{\text{MFC}}/0/90/90^{\text{MFC}}]$ laminate, and the experimental equipment is shown in Fig. 13. Some samples are prepared by autoclave processing using T700 prepreg. MFC($85 \text{ mm} \times 57 \text{ mm}$) were bonded to both sides of the laminate using LOCTITE epoxy resin adhesive E-120HP. The MFC is connected to a piezoelectric controller and a computer sends a signal to apply a voltage excitation. The midpoint of the laminate is fixed on the top of the vibration exciter, and the non-contact laser displacement sensor is installed above the laminate and connected with the data acquisition board to measure the displacement.

The four curves in Fig. 14 represent the quasi-static forward loading and backward loading processes in the experimental and analytical model respectively, and the arrows indicate the direction of the snap-through. The red line represents the enabled MFC1 to achieve the snap-through from stable state A to stable state B. By applying a positive voltage, MFC1 extends to reduce the deflection of the bistable laminate. When the critical voltage point is reached, the vertex displacement undergoes a sudden change and transition to a value opposite to the previous sign. The voltage at this abrupt point is the quasi-static critical snap-through voltage, which is 3180 V under the forward loading of the analytical model. The yellow line represents the forward loading in the experiment, under the same voltage that was applied to MFC1. Due to the power and working voltage limitations of the MFC (maximum voltage 1500 V), and the stable state A fails to snap through, but the deformation trend is similar to the prediction result given by the analytical model. The blue line indicates that MFC2 is enabled to achieve snap-through from stable state B to stable state A. The critical snap-through voltage of the backward loading in the analytical model is 425 V, while the experimental value is 750 V, which is slightly larger than the analytical result. The analytical model idealizes the connection between the MFC and the bistable laminate as perfect based on the Kirchhoff hypothesis. However, the actual adhesive bond in experiments has measurable elasticity and thickness, which introduces a compliance effect. This imperfect transfer of strain acts as a constraint that is not faithfully replicated in the model, ultimately causing the deviation in results.

According to the result predicted by the force-displacement curve, the snap-through from stable state B to stable state A can be easily realized. This indicates that the analytical model not only can accurately predict the stable configuration but also is able to analyse the snap-through process. During the snap-through process, the maximum prediction error of the displacement is about 17 %, which mainly comes from the hypothesis of theoretical model and manufacturing errors. It is foreseeable that stable configurations bending along the short side are easier to actuate than stable configurations bending along the long side, so the subsequent optimization mainly focuses on the actuation of stable configurations bending along the long side.

3.3. Dynamic analysis of bistable laminate

When the excitation load remains unchanged, the displacement of the structure is the largest at the resonant frequency. It can be foreseen that the larger the displacement, the easier it is to achieve snap-through. Subsequent research can prove that resonant actuation is the optimal solution for dynamic actuation. In order to demonstrate the nonlinear characteristics of the bistable laminate specimens under sinu-

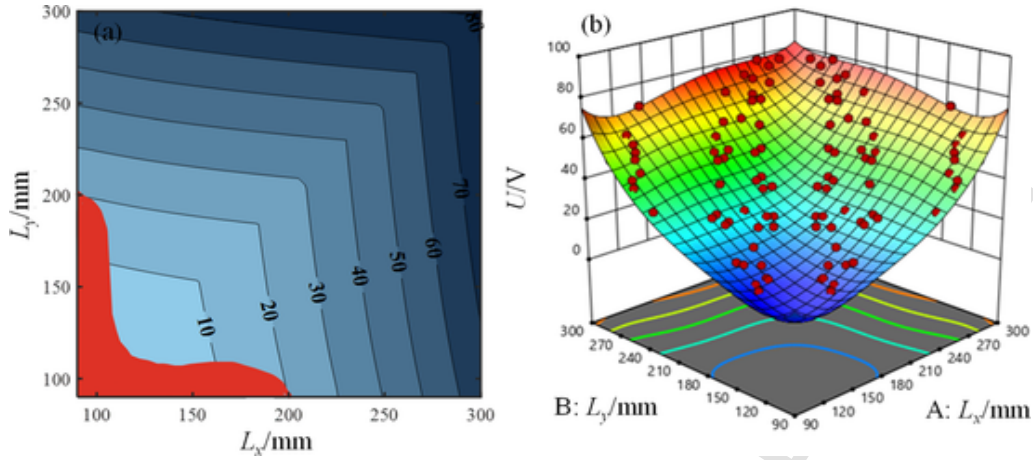


Fig. 30. The deflections of MFC-actuated bistable laminates, (a) two-dimensional contour plot, (b) three-dimensional plot.

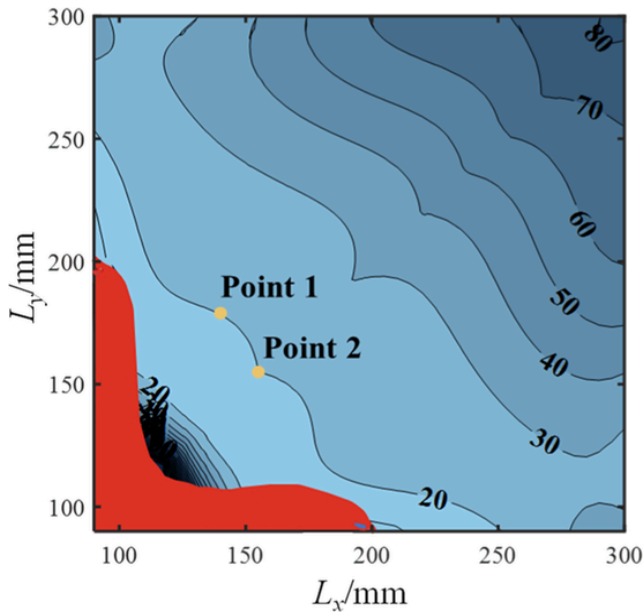


Fig. 31. The actuation voltage required for one unit deflection of the bistable laminate.

soidal excitation, frequency sweep testing of the piezoelectric bistable laminates is carried out using the equipment shown in Fig. 13. The vibration exciter is connected to the vibration controller through a power amplifier. An acceleration sensor was integrated into this closed-loop system for sinusoidal frequency sweep testing. The response of the structure under different intensities of excitation is recorded by using laser displacement sensor. The frequency range of the excitation is from 20 Hz to 60 Hz, and the amplitude of the excitation is from 1 g to 6 g ($g = 9.8 \text{ m/s}^2$), sweeping from low frequency to high frequency and then from high frequency to low frequency respectively.

As a typical nonlinear structure, the natural frequency of bistable laminate varies with the change of external excitation. To facilitate the subsequent study of resonant actuation, the backbone curve is introduced to visualize and quantify this nonlinear behaviour. The backbone curve represents the trajectory of the system's resonant frequency varying with the amplitude. The frequency response curves of bistable piezoelectric composite laminate under different strengths of harmonic excitation under forward and backward loading are obtained, and the

resonant peaks of the frequency response curves are further connected to extract the backbone curve as shown in Fig. 15.

The frequency response curves of the forward sweep frequency and backward sweep frequency vibration testing at excitation amplitudes from 1 g to 6 g ($g = 9.8 \text{ m/s}^2$) are obtained for the bistable laminates with MFC. As shown in Fig. 15, the solid and dotted lines represent the forward and backward sweep frequency testing, respectively. When the external excitation intensity is relatively low, the frequency response curve of the structure is approximately symmetric and presents linear characteristics. With the increase of the applied excitation intensity, the resonant peaks in the frequency response curve begin to shift to the low frequency. The resonant frequency of the structure decreases with the increase of the applied excitation intensity, showing the soft spring characteristic. With the increase of excitation amplitude, the resonant frequency of the structure gradually decreases from 38.4 Hz to 37.1 Hz, and the stiffness of the system gradually decreases with the increase of excitation. Hysteresis phenomenon exists between forward and backward sweep frequency testings, and the system has soft dynamic hysteresis. Hysteresis and snap-through are important characteristics of nonlinear forced vibration systems. The soft dynamic hysteresis of the structure also imply that the structure has multi-value responses and initial value sensitivity. The final response of the structure not only depends on the excitation but is also affected by the initial conditions and historical states. The addition of MFC broadens the resonant region of the structure while reducing the response amplitude of the stable state. However, for the same bistable laminate, the resonance regions of the structure are different under different actuation intensities. In order to find the critical voltage for the structure under resonant actuation, the following text combines the Poincaré map to accurately determine the resonance region of the structure under the critical voltage, so as to determine the critical voltage for resonant actuation.

3.4. Resonant actuation and snap-through modes

Because the resonant peaks of the structure change at different excitations, it is difficult to directly find the critical load of the resonant actuation. The bistable laminates of different sizes have different stable state responses under different excitations, and the snap-through modes of the bistable laminates are first investigated. Apply sinusoidal excitation of different amplitudes and frequencies to the structure for a relatively long period of time, usually 20 s, and extract a typical response that reflects the characteristics of the structure. By identifying the phase diagrams and Poincaré map of the stable state responses, the snap-through modes of the bistable laminates are classified into three main categories: transient cross-well oscillation, continuous cross-well oscil-

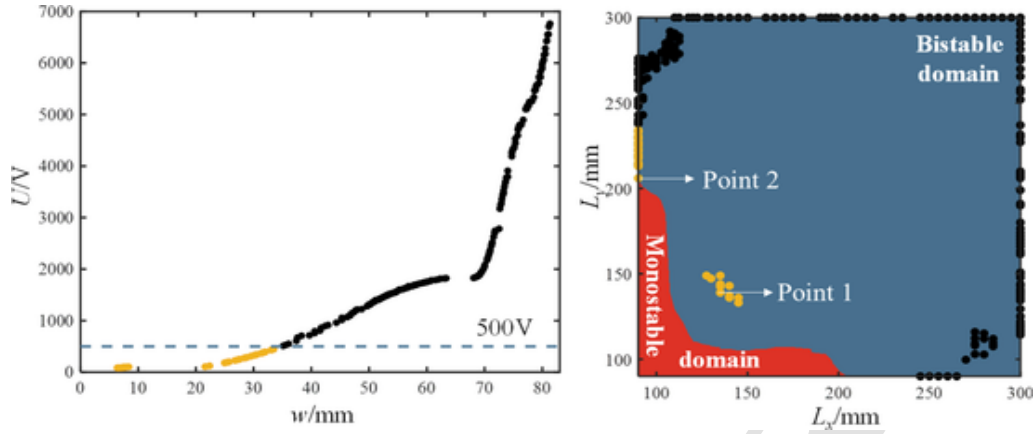


Fig. 32. Pareto optimality, (a) Pareto front, (b) the distribution of Pareto set.

lations and chaotic oscillation. It represents that the structure snaps through under the critical resonant actuation, corresponding to the transient snap-through, continuous snap-through and chaotic snap-through, respectively. Figs. 16–18 show the displacement responses, phase diagrams, Poincaré maps and spectrum maps of different structures, which only consider the stable state response after the transient responses are exhausted. Figs. 16(a), 17(a) and 18(a) show the typical time-domain displacement curves in one second under different excitation loads. The red dotted line and the blue dotted line represent the stable positions of stable state A and stable state B respectively. In Fig. 16(a), the response mode of the structure can be clearly seen from the curve, while it is difficult to distinguish the response mode of the structure from Figs. 17(a) and 18(a). Then further analysis through phase diagrams and spectrum maps is required.

When the external energy input is insufficient, the structure oscillates within a single potential energy well and is unable to cross the energy barrier to achieve snap-through. When the excitation intensity is sufficient, the bistable laminates begin to snap through. As shown in Fig. 16, the transient cross-well oscillation is quasi-periodic vibration before the snap-through, chaotic oscillation occurs during the snap-through, and periodic oscillation occurs after the snap-through to the other stable configuration. Continuous cross-well oscillation is shown in Fig. 17, where the structure snaps through and does regular cross-well oscillations in the two potential energy wells. Continuous cross-well motion involves the simultaneous occurrence of quasi-periodic and chaotic oscillations, which periodic response are interrupted by the non-periodic response. The Poincaré map of the stable state response of quasi-periodic oscillation is multiple isolated clusters of points. Chaotic snap-through as shown in Fig. 18, the structure experienced two stable configurations doing irregular motions, and it is determined that the structure undergoes chaos through phase portrait and Poincaré map.

For each spectrum map, the main frequency of structural vibration is the excitation frequency. For single-well oscillation and transient cross-well oscillation, the structure also shows periodic doubling oscillation. For chaotic motion, there is a continuous spectrum in other regions except the main peak. Multiple points in a Poincaré map correspond to multiple peaks in a spectrum map.

In order to find the critical load of the resonant actuation, the bifurcation diagram is used to determine the critical load amplitude of different snap-through modes. A bifurcation diagram is used to visualize the structure nonlinear by applying different excitations to the structure. Taking a 200 mm × 100 mm bistable piezoelectric laminate as an example, the critical loads of two stable configurations by resonant actuation are obtained. Figs. 19 and 20 respectively. It shows the bifurcation diagram of stable state response of stable configuration A under different intensity voltages. The left diagram is the displacement bifur-

cation diagram, and the right diagram is the displacement time history response corresponding to the red line. First, the resonant frequency of the stable state A (Fig. 19) is identified. The critical excitation amplitude of the resonant actuation given in Fig. 20, is determined.

As shown in Fig. 19(a), sinusoidal voltages of different intensities are applied to MFC 1 on the stable state A, where U and Ω represent the amplitude and frequency of the sinusoidal excitation respectively. Fixing amplitude invariant, and the excitation frequency Ω gradually increases from 40 Hz to 60 Hz. When the amplitude U is 320 V, the response is only periodic oscillation and quasi-periodic oscillation, and the structure does not occur snap-through. Fig. 19(b) represents the displacement time history under excitation corresponding to the red vertical line in the left figure. It can be seen that even in the resonant region, the structure does not snap through. When the voltage amplitude is 330 V, as shown in Fig. 19(c), the response begins to diverge and the structure undergoes a chaotic snap-through. With the further increase of voltage amplitude to 340 V, the chaotic region of the structure is gradually widened. In the region outside the chaos, the structure oscillates periodically around the stable state A. When the resonance region is reached, the stable state response of the structure enters chaos and reaches the stable state B in the process.

It can be determined from Fig. 19 that the critical snap-through frequency of stable state A is 51.2 Hz. Then the excitation frequency remains constant at 51.2 Hz, and the excitation amplitude voltage gradually increases from 200 V to 500 V, and the bifurcation diagram of the displacement response of the stable state A is shown in Fig. 20. It can be seen, from Fig. 20, with the increase of excitation amplitude, the motion state of the structure changes from single-well periodic oscillation to quasi-periodic oscillation, and then to cross-well chaotic oscillation, eventually leading to snaps through in the chaotic oscillation. It is successfully found that the critical voltage amplitude and frequency of stable state A to realize the snap-through under resonant actuation is 329 V and 51.2 Hz.

The critical voltage amplitude and corresponding excitation frequency of stable state B are determined using the same method. Similarly, when the amplitude is 160 V, the structure does not snap through and always oscillates periodically around the stable state B as shown in Fig. 21(a). When the amplitude reaches 170 V, the displacement bifurcation of the structure is shown in Fig. 21(c). The structure first oscillates periodically near the stable state B. After a period of loading time, the structure undergoes a snap-through near the resonant region and continues to oscillate periodically near the stable state A. As the voltage amplitude increases, the bandwidth within which the snap-through can occur also increases.

It can be determined from Fig. 21 that the resonant frequency of stable state B when the snap-through occurs is about 46.1 Hz. With the fre-

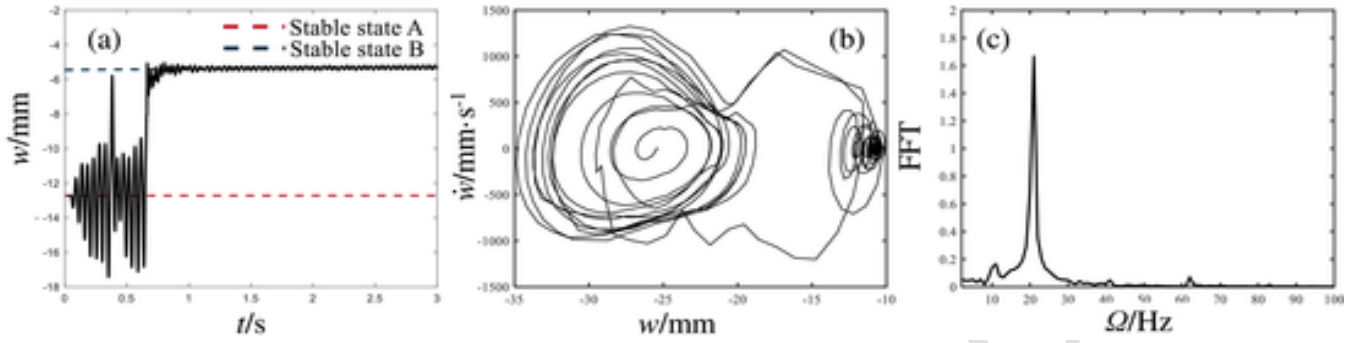


Fig. 33. The snap-through of Point 1, (a) displacement curve, (b) phase diagram, (c) spectrum map.

quency unchanged, the excitation amplitude voltage is gradually increased from 100 V to 400 V. The bifurcation diagram of the displacement response of the stable state B is shown in Fig. 22. As shown in Fig. 22, different from the response of stable state A, the bifurcation diagram of stable state B has two different states. Different with the forward loading, stable state B becomes chaotic only when the snap-through occurs. It moves periodically around stable state B and stable state A before and after the snap-through. It can be determined that the critical voltage amplitude and frequency of stable state B to achieve a snap-through under resonant actuation are 170 V and 46.1 Hz, respectively. The forward snap-through and backward snap-through of 200 mm × 100 mm bistable laminates correspond to chaotic snap-through and transient cross-well snap-through, respectively.

Under forward loading, MFC1 has an actuating voltage amplitude of 329 V and a resonant frequency of 51.2 Hz. When applying the backward loading, the MFC2 has an actuating voltage amplitude of 170 V and a resonant frequency of 46.5 Hz. The voltage required for resonant actuation is much lower than that of the quasi-static actuation, which is 3180 V and 425 V. Resonant actuation can greatly reduce the voltage required for snap-through. The experimental deformation trend is consistent with that predicted by the analytical model, but the voltage required by the experiment is larger than that of the analytical model. For the more compliant stable state A, the resonant actuation greatly reduces the voltage required to snap through.

4. Multi-objective optimization and experiment verification

Considering the application of bistable laminates on morphing wings, it is expected to find the bistable laminates that meet the requirements. With the length and width of the bistable laminates as design variables, the responses are critical actuation voltage and vertex deflection. The bistable laminates with large deformation and low actuation requirements will be found using an optimization process. The multi-objective optimization design model of bistable laminates is shown in Eq. (29), with two optimization objectives and three constraints. By establishing the relationship between the output variables and multiple independent variables through the RSM, the critical actuation voltage and deflection of the bistable laminates within the entire variable range can be obtained with limited calculations, providing a model for subsequent optimization. In this paper, the software Design-Expert is used to implement the response surface method. In the software Design-Expert, the global distribution of actuation voltage and vertex deflection is obtained by fitting the sixth-order model with the central composite design (CCD) of RSM. The model is verified by ANOVA after the fitting, and an accurate surrogate model is obtained. Then NSGA-II was used to search for the Pareto front in the region, and two groups of specimens are selected for experimental verification. The whole process mainly includes experimental design, theoretical calcu-

lation, RSM fitting, finding Pareto front. Two sets of optimal solutions are selected in the Pareto front and verified in the experiments.

$$\text{Opt.} \begin{cases} \text{Max : } w \\ \text{Min : } U \\ \text{s.t. : } U \leq 500V \\ L_x \in [90, 300] \\ L_y \in [90, 300] \end{cases} \quad (29)$$

where w is vertex deflection. Because the potential energy of stable state A is lower than that of stable state B, the structure is more likely to escape from the potential energy well of stable state B. The subsequent optimization of the structure mainly focuses on the forward actuation of the laminate, from stable state A to stable state B. U represents the critical load amplitude of forward actuation.

4.1. Sampling of training data for the surrogate model

Using a 85 mm × 57 mm MFC as the actuator, the length and width of the laminates range from 90 mm to 300 mm. The cured and cooled configurations of 22,366 laminates with dimensions varying from 90 mm to 300 mm (in 1 mm increments) were systematically evaluated. Firstly, after obtaining the deflection of the bistable laminates in the entire area through the RSM, it is necessary to determine the monostable domain and bistable domain of the composite laminates in this region through the method described in Section 2.2. The bistability of the laminate is judged by the number of structural potential energy wells. Then plot the discrete results as a continuous surface, as shown in Fig. 23. The laminates in the red region have only one potential energy well and only one stable configuration. The asymmetric composite laminates in the blue area are bistable structures with two potential energy wells. The quantification of the energy barrier is not able to be realized directly because of many factors, which is particularly important for selecting a suitable actuation mechanism.

Fig. 24 shows that 100 points are selected as the training set by the optimal Latin hypercube design in the region from 90 mm to 300 mm, and it can be seen that the spatial distribution of samples is relatively uniform and reasonable. After determining the bistable region and the training set, the critical resonant actuation amplitude of the training set is calculated by the method in Chapter 3. As shown in Fig. 24, the blue dots near the monostable domain indicate that the structure mainly behaves as a chaotic snap-through under resonant actuation. Since one of the stable configurations of the structure near the monostable region has a much lower energy barrier than the other stable state. Even if the resonant frequencies of the stable configurations are not close to each other, the energy input that allows the structure to cross a large energy barrier is able to enable the structure

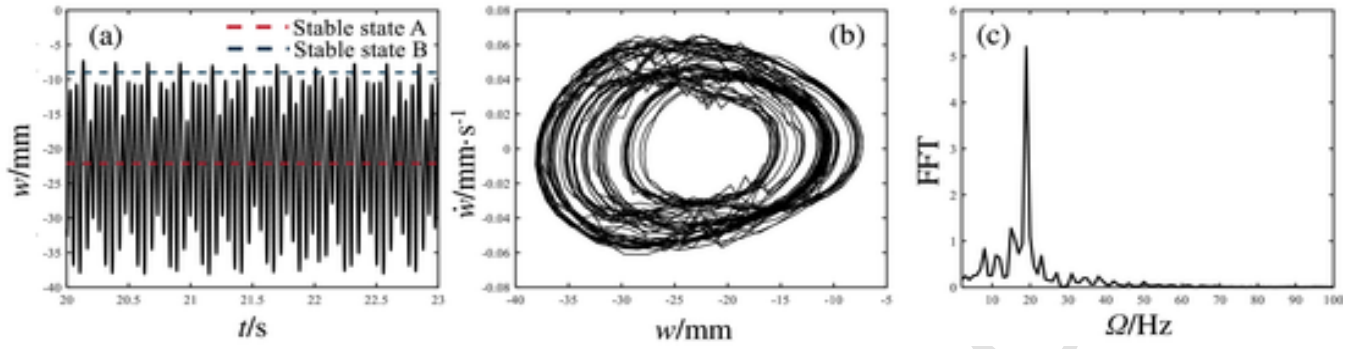


Fig. 34. The snap-through of Point 2, (a) displacement curve, (b) phase diagram, (c) spectrum map.

to easily escape from the shallower potential energy well. The red dots in the middle region represent that the snap-through form of the bistable laminate is transient cross-well oscillation. The yellow dots represent the snap-through form of the structure with both continuous and transient cross-well oscillations. In the region studied in this paper, most of the structures are snap-throughs by transient cross-well motion.

4.2. The surrogate model of RSM

After obtaining the critical actuation voltage and the deflection of the stable state A for all training points, the data are entered into the software Design-Expert 13. In this paper, the models suitable for both actuation load and deflection are sixth-order. The multivariate correlation coefficient R^2 greater than 0.9 approves that the model is reasonable, with higher values indicating better correlation. In the model of critical actuation voltage, R^2 is 0.9925. The values of Adj R-Squared and Pred R-Squared need to be high and close, and the results in this paper are 0.9912 and 0.9885, respectively, which indicates that the surrogate model can explain this process well. Adeq Prediction is the ratio of effective signal to noise, > 4 is considered reasonable, and this model is 102.4. In the cross-validation error analysis, the actual results of the analytical model and predicted results of the surrogate model are shown in Fig. 25. As shown in Fig. 25(a), the residuals are linearly normal distribution, indicating that the model fits well. In Fig. 25(b), the distribution of residual and predicted values is mainly irregular. Fig. 25(c) shows that the distribution of predicted and measured values is mostly near the diagonal, which proves that the predicted results of RSM are acceptable.

In order to further verify the accuracy of the surrogate model, 50 points are randomly selected as the validation set in addition to the original 100 training points, as shown in Fig. 26(a). Compared the prediction results of the surrogate model with the analytical model, as shown in Fig. 26(b). The results are relatively similar in most regions, and only a few regions have large errors. Among the 50 verification points, the maximum error is 12.31 %, the minimum error is 0.022 %, and the average error is 4.89 % which is about 37.85 V. From the 100 training points and 50 verification points, the surrogate model is well consistent with the actual calculation results of the analytical model, and the model can be used in the subsequent analysis.

Fig. 27 shows the critical voltage distribution of resonant actuation predicted by the RSM. It can be seen from Fig. 27 that there is no linear relationship between the amplitude of the voltage and the size or area of the laminate. In general, larger the bistable laminate tends to have smaller the resonant frequency, larger deflection, and higher required actuation voltage. From stable state A to stable state B, the actuation voltage of square laminates is higher than that of rectangular laminates

with the same area level of laminate. The range of actuation voltages in this region spans from 110 V to 6840 V.

The variables are still the length and width of the laminate, and the response R2 is the vertex deflection of the stable state A of the bistable laminate. For the surrogate model of deflection, R^2 is 0.9983, Adj R-Squared and Pred R-Squared have values of 0.9980 and 0.9972, and Adeq Prediction is 202.8203. The predictions of the surrogate model and the analytic model about the results are shown in Fig. 28. Similarly, the model is qualified.

The 50 points in Fig. 26(a) are also taken to further verify the accuracy of RSM's prediction of maximum deflection. The vertex deflections of the prediction results of the analytical model and RSM are shown in Fig. 29, and the results are similar in most areas. In 50 verification sets, the maximum error is 6.71 %, the minimum error is 0.62 %, the average error is 1.85 %, and the average error of all data is 0.665 mm. The prediction error of the RSM model is smaller.

The predicted results of deflection are shown in Fig. 30, and there is nonlinear relationship between deflection and the size or area of the laminate. Overall, the larger the bistable laminates, the greater the deflection. Regarding stable state A, the deflection of square laminates is larger than that of rectangular laminates with the same area. The deflection spans from 7.3779 mm to 78.3689 mm over the region.

4.3. Multi-objective optimization of RSM

NSGA-II is used to obtain the non-dominated solution of the system, that is, the Pareto optimality. In order to obtain a bistable laminate with small critical actuation voltage and large deflection, the objective function of the Pareto optimality is the actuation voltage required per unit deflection. By dividing the corresponding actuation voltage by the vertex deflection, the actuation voltage required per unit deflection is obtained, and the calculation results are shown in Fig. 31. Structures that not only meet the required deformation in applications but also achieve the goal of low-voltage actuation are called optimal structures. The voltage actuation range of the MFC is from -500 V to 1500 V. Exceeding this voltage may cause permanent damage to the material. Therefore, taking 500 V as the limit, the approximate set of the Pareto front is divided into two sets of yellow and black points by limiting the voltage to be < 500 V, as shown in Fig. 32(a). In Fig. 32(b), it is observed the yellow point set is mainly distributed in two regions, so a point with the lowest actuation voltage is selected from each of these two regions. Verify through experiments whether resonance actuation can be accomplished at low voltage. The dimensions of Point 1 are 140 mm $\times$ 135 mm, and the dimensions of Point 2 are 205 mm $\times$ 90 mm.

4.4. Experimental verification

As shown in Fig. 14, certain deviations exist between the analytical model predictions and the experimental results. Therefore, the outcomes of the multi-objective optimization, which used the analytical data to train a surrogate model, require further experimental verification—specifically to confirm the determination of bistable laminate geometric dimensions that combine large deformation with low actuation voltage.

For the 85 mm × 57 mm MFC, two bistable laminates are randomly selected for experimental verification after optimization. The sizes of the bistable laminated plates are 140 mm × 135 mm and 205 mm × 90 mm. The calculation results of the analytical model indicate that the vertex deflection of the 140 mm × 135 mm laminate is 11.9 mm, and the amplitude of the resonant actuation voltage is 238 V. The vertex deflection of the 205 mm × 90 mm laminate is 17.2 mm, and the amplitude of the resonant actuation voltage is 344 V. Since it is difficult to realize the snap-through in the experiment under power constraints, two 5.42 g inertial masses are added to the left and right ends of the bistable laminates to increase the inertial force and reduce the resonant frequency. Fig. 33 shows the displacement time response, phase diagram and spectrum map of Point 1 under $U = 500$ V, $\Omega = 19$ Hz forced excitation. The bistable laminate has successfully realized the snap-through after adding inertial masses. In the motion process, the structure first experiences quasi-periodic oscillation, then snaps through to the stable state B and continues to oscillate with a small amplitude. After the excitation stops, the laminates remain in stable state B.

For Point 2, the applied load is $U = 500$ V, $\Omega = 21$ Hz, and the response is shown in Fig. 34. During the experiment, the bistable laminate makes a unique “popping” sound when snapping through, which indicates that the structure has snapped through. The structure does continuous cross-well motion under forced excitation and can't stay in the stable state A or stable state B for periodic oscillations.

5. Conclusions

Although the advantages of multistable structures are widely acknowledged, ongoing interest remains in further understanding the physical mechanisms of snap-through and in developing efficient actuation methods. From the perspective of structural dynamics and control, multi-stable structures represent mechanically guided variable-stiffness flexible systems with strong nonlinearity such as large deformations and bifurcations. There exists a complex coupling relationship between the mechanical properties of multi-stable structures and the actuation systems. The theoretical foundations for the design of stable configurations, switching path planning, vibration bandwidth adjustment, and low-energy consumption actuation of multi-stable structure systems require further development. This paper focuses on the research of these scientific issues and technical challenges. Based on theory, simulation and experiment, this paper proposes and verifies the feasibility of resonant actuation from static and dynamic analysis. The findings can be used to enhance existing theories on the dynamics of bistable structures and provide a theoretical basis for the use of new concept morphing structures with similar mechanical properties.

After comparing the fourth-order and sixth-order analytical models, precise stable configurations are obtained. A systematic analysis and comparison are conducted on parameters such as dimensions of the laminate, aspect ratio, adhesive layer thickness, and concentrated load. Under the condition of a fixed central boundary, compared with finite element model, the sixth-order model showed better consistency in the large deflection structural region, while the fourth-order model exhibited better convergence. Both experimental and theoretical results indicate that it is challenging for a bistable laminate with large deflection to undergo a snap-through under limits.

For bistable laminates of different sizes, under different intensity and frequency harmonic excitation, there are three different snap-through modes: transient snap-through, continuous snap-through and chaotic snap-through. Poincaré maps are used to visualize the critical actuation conditions for the different snap-through modes. Both experimental and theoretical results show that the voltage required for the resonant actuation is much smaller than that for quasi-static actuation.

Considering the dimensions of the composite laminate as variables, resonant actuation critical voltage amplitudes are calculated at appropriate test points. This paper proposed a resonant actuation strategy and detailed a method for accurately determining the critical actuation voltage for such nonlinear structures. By dividing the region based on different critical snap-through modes in resonant actuation, a global surrogate model is obtained using the sixth-order RSM to reduce the computational costs. By taking into account the requirements of significant deformation and ease of actuation, find the Pareto front of the actuation voltage required per unit deflection. The optimal solution is selected and verified by experiment. The low energy path exploration and global optimization of MFC actuation are completed, which lays a theoretical foundation for its application in morphing wings.

CRediT authorship contribution statement

Yuting liu: Writing – original draft. **Ke Huang:** Resources, Investigation. **Jiaying Zhang:** Writing – review & editing, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This project has received funding from National Natural Science Foundation of China (Grant No. 92271104, 12102017) and Beijing Natural Science Foundation (Grant No. 1232014).

Data availability

No data was used for the research described in the article.

References

- [1] D. Bashir, P.M. Anilkumar, S. Scheffler, A. Haldar, B.N. Rao, R. Rolfes, A review of the dynamic behavior of thermally induced bistable configurations of unsymmetrical composite laminates and their applications, *Arch. Comput. Methods Eng.* 32 (2025) 1635–1677.
- [2] C. Zhang, X.F. Yin, R.C. Chen, K.W. Ju, Y.X. Hao, T. Wu, J. Sun, H.X. Yang, Y. Xu, A review on reprogrammable bistable structures, *Smart Mater. Struct.* 33 (2024).
- [3] D.M. Lemos, F.D. Marques, A.J.M. Ferreira, A review on bistable composite laminates for aerospace applications, *Compos. Struct.* 329 (2024) 117756.
- [4] S.A. Emam, D.J. Inman, A review on bistable composite laminates for morphing and energy harvesting, *Appl. Mech. Rev.* 67 (6) (2015) 060803.
- [5] M.W. Hyer, Some observations on the cured shape of thin unsymmetric laminates, *J. Compos. Mater.* 15 (2) (1981) 175–194.
- [6] M.W. Hyer, The room-temperature shapes of four-layer unsymmetric cross-ply laminates, *J. Compos. Mater.* 16 (1982) 318–340.
- [7] M.L. Dano, M.W. Hyer, Thermally-induced deformation behavior of unsymmetric laminates, *Int. J. Solids Struct.* 35 (1998) 2101–2120.
- [8] B.D. Harper, The effects of moisture induced swelling upon the shapes of anti-symmetric cross-ply laminates, *J. Compos. Mater.* 21 (1987) 36–48.
- [9] W.J. Jun, C.S. Hong, Effect of residual shear strain on the cured shape of unsymmetric cross-ply thin laminates, *Compos. Sci. Technol.* 38 (1990) 55–67.
- [10] W.J. Jun, C.S. Hong, Cured shape of unsymmetric laminates with arbitrary lay-up angles, *J. Reinf. Plast. Compos.* 11 (12) (1991) 1352–1366.
- [11] M. Schlecht, K. Schulte, Advanced calculation of the room-temperature shapes of unsymmetric laminates, *J. Compos. Mater.* 33 (16) (1995) 1472–1490.
- [12] L.J.B. Peeters, P.C. Powell, L. Warnet, Thermally-induced shapes of unsymmetric laminates, *J. Compos. Mater.* 30 (5) (1996) 603–626.
- [13] F. Mattioni, P. Weaver, M. Friswell, K. Potter, Modeling and application of

- thermally induced multistable composites with piece-wise variation of lay-up in the planform. 15th AIAA/ASME/AHS Adaptive Structures Conference. AIAA paper No. 2007-2264, 2007.
- [14] Z. Wu, H. Li, M.I. Friswell, Advanced nonlinear dynamic modelling of bi-stable composite plates, *Compos. Struct.* 201 (2018) 582–596.
 - [15] A. Pandey, A. Arockiarajan, Actuation performance of macro-fiber composite (MFC): modeling and experimental studies, *Sens. Actuators A-Phys.* 248 (2016) 114–129.
 - [16] M. Dano, M.W. Hyer, Snap-through of unsymmetric fiber-reinforced composite laminates, *Int. J. Solids Struct.* 39 (2002) 175–198.
 - [17] M. Dano, M.W. Hyer, SMA-induced snap-through of unsymmetric fiber-reinforced composite laminates, *Int. J. Solids Struct.* 40 (2003) 5949–5972.
 - [18] A. Pirondi, Y. Gandhi, L. Collini, Strategies for modelling and optimization of bi-stable composite laminates actuated by embedded Shape Memory Alloy wires, *Procedia Struct. Integr.* (2019) 455–469.
 - [19] M.R. Schultz, M.W. Hyer, Snap-through of unsymmetric cross-ply laminates using piezoceramic actuators, *J. Intell. Mater. Syst. Struct.* 14 (12) (2003) 795–814.
 - [20] M.R. Schultz, R. Marc, M.W. Hyer, A morphing concept based on unsymmetric composite laminates and piezoceramic MFC actuators, in: 45th AIAA/ASME/ASCE/AHS/ASC structures, structural dynamics and materials conference. AIAA paper No. 2004-1806, 2004.
 - [21] M.R. Schultz, W.K. Wilkie, R. Bryant, Investigation of self-resetting active multistable laminates, *J. Aircr.* 44 (2005) 1069–1076.
 - [22] C. Bowen, R. Butler, R. Jervis, H. Kim, A. Salo, Morphing and shape control using unsymmetrical composites, *J. Intell. Mater. Syst. Struct.* 18 (1) (2007) 89–98.
 - [23] P. Portela, P. Camanho, P. Weaver, I. Bond, Analysis of morphing, multi stable structures actuated by piezoelectric patches, *Comput. Struct.* 86 (2008) 347–356.
 - [24] H.A. Kim, D.N. Betts, A.I.T. Salo, Shape memory alloy–piezoelectric active structures for reversible actuation of bistable composites, *AIAA J.* 48 (6) (2010) 1265–1268.
 - [25] A.J. Lee, A. Moosavian, D.J. Inman, Control and characterization of a bistable laminate generated with piezoelectricity, *Smart Mater. Struct.* 26 (8) (2017) 085007.
 - [26] J. Zhang, Z. Wu, C. Zhang, L. Hao, R. Nie, J. Qiu, Nonlinear dynamics of shape memory alloys actuated bistable beams, *Smart Mater. Struct.* 28 (2019) 055009.
 - [27] P.M. Anilkumar, A. Haldar, S. Scheffler, E. Jansen, B. Rao, R. Rolfes, Morphing of bistable variable stiffness composites using distributed MFC actuators, *Compos. Struct.* 289 (2022) 115396.
 - [28] V. Deshpande, O.J. Myers, G.M. Fadel, S. Li, Transient deformation and curvature evolution during the snap-through of a bistable laminate under asymmetric point load, *Compos. Sci. Technol.* 211 (2021) 108871.
 - [29] J. Casals-Terre, A. Fargas-Marques, A.M. Shkel, Snap-action bistable micromechanisms actuated by nonlinear resonance, *J. Microelectromechanical Syst.* 17 (5) (2008) 1082–1093.
 - [30] C.G. Diaconu, P.M. Weaver, A.F. Arrieta, Dynamic analysis of bi-stable composite plates, *J. Sound Vib.* 322 (2009) 987–1004.
 - [31] A. Senba, T. Ikeda, T. Ueda, A two-way morphing actuation of bi-stable composites with piezoelectric fibers, *Proceedings of the 51st AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics, and Materials Conference 18th AIAA/ASME/AHS Adaptive Structures Conference*, AIAA paper No. 2010-2744, 2010.
 - [32] J. Wang, P. Zhang, X. Li, et al., Nonlinear dynamics of bistable rectangular asymmetric composite lamina-te: theory, simulation and experiment, *Appl. Math. Model.* (2025) 116183.
 - [33] P. Zhang, X. Li, J. Wang, Nonlinear effect on stable state and snap-through bistability of square composite laminate, *Eur. J. Mech.-A/Solids* 109 (2025) 105431.
 - [34] A.F. Arrieta, S.A. Neild, D.J. Wagg, Nonlinear dynamic response and modeling of a bi-stable composite plate for applications to adaptive structures, *Nonlinear Dyn.* 58 (2009) 259–272.
 - [35] A.F. Arrieta, O. Bilgen, M.I. Friswell, P. Hagedorn, Dynamic control for morphing of bi-stable composites, *J. Intell. Mater. Syst. Struct.* 24 (3) (2013) 266–273.
 - [36] O. Bilgen, A. Arrieta, M. Friswell, P. Hagedorn, Dynamic control of a bistable wing under aerodynamic loading, *Smart Mater. Struct.* 22 (2013) 025020.
 - [37] A.F. Arrieta, O. Bilgen, M.I. Friswell, Modelling and configuration control of wing-shaped bi-stable piezoelectric composites under aerodynamic loads, *Aerosp. Sci. Technol.* 29 (2013) 453–461.
 - [38] M. Rouleau, S.R. Craig, Y. Xia, R. Shieh, M.L. Robinson, C. Shi, J. Meaud, Nonlinear dynamics of a bistable system impacting a sinusoidally vibrating shaker, *Nonlinear Dyn.* 110 (2022) 3015–3030.
 - [39] M.Q. Wu, W. Zhang, D. Chronopoulos, Nonlinear vibrations and dynamic snap-through behaviors of four-corner simply supported bistable asymmetric laminated composite square shell, *Mech. Syst. Sig. Process* 173 (2022) 109023.
 - [40] A.A. Rader, F.F. Afagh, A. Yousefi-Koma, D.G. Zimcik, Optimization of piezoelectric actuator configuration on a flexible fin for vibration control using genetic algorithms, *J. Intell. Mater. Syst. Struct.* 18 (10) (2007) 1015–1033.
 - [41] C. Diaconu, P. Weaver, F. Mattioni, Concepts for morphing airfoil sections using bi-stable laminated composite structures, *Thin-Walled Struct.* 46 (2008) 689–701.
 - [42] A.S. Panesar, P.M. Weaver, Optimisation of blended bistable laminates for a morphing flap, *Compos. Struct.* 94 (2012) 3092–3105.
 - [43] D.N. Betts, H.A. Kim, C.R. Bowen, Modeling and optimization of bistable composite laminates for piezoelectric actuation, *J. Intell. Mater. Syst. Struct.* 22 (18) (2011) 2181–2191.
 - [44] F. Nicassio, G. Scarselli, F. Pinto, F. Ciampa, O. Iervolino, M. Meo, Low energy actuation technique of bistable composites for aircraft morphing, *Aerosp. Sci. Technol.* 75 (2018) 35–46.
 - [45] Z. Zhang, X. Li, X. Yu, Magnetic actuation bionic robotic gripper with bistable morphing structure, *Compos. Struct.* 229 (2019) 111422.
 - [46] X. Wang, K. Aarjav, C. Chao, A soft pneumatic bistable reinforced actuator bioinspired by Venus Flytrap with enhanced grasping capability, *Bioinspir. Biomim.* 15 (2020) 056017.
 - [47] Z. Lu, D. Shao, Z. Fang, H. Ding, L. Chen, Integrated vibration isolation and energy harvesting via a bistable piezo-composite plate, *J. Vib. Control* 26 (2019) 779–789.
 - [48] N. Vasios, B. Deng, B. Gorissen, K. Bertoldi, Universally bistable shells with nonzero gaussian curvature for two-way transition waves, *Nat. Commun.* 12 (1) (2021) 1–9.
 - [49] T. Dong, W. Zhang, M. Dong, The novel morphing airfoil based on the bistable composite laminated shell, *Nonlinear Dyn.* 111 (19) (2023) 17667–17685.
 - [50] T. Dong, Y. Chen, M. Dong, et al., The intra-well and cross-well dynamics of the unsymmetric bistable composite laminated shells, *Nonlinear Dyn.* 113 (2025) 27289–27322.
 - [51] D. Bashir, P.M. Anikumar, B.N. Rao, Exploring the potential energy landscape of square bistable laminates bonded with piezoelectric macro fiber composites, *Compos. Struct.* 369 (2025) 119239.
 - [52] A.F. Arrieta, V.V. Gemmeren, A.J. Anderson, P.M. Weaver, Dynamics and control of twisting bi-stable structures, *Smart Mater. Struct.* 27 (2018).